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Digital twin with Machine learning for predictive monitoring of CO2 equivalent from existing buildings

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Digital Twin with Machine Learning for Predictive Monitoring of CO_2 equivalent from Existing Buildings

Abstract

The revolution of the industry 4.0 presents a new era of digital transformation for the construction industry, advancing towards the concept of digital twins, while on the other hand it faces the global challenge of reducing carbon emissions from operational assets. The current research gap in the application of digital twins for achieving net zero were reviewed in this study, highlighting its potentials for enhancement in the built environment, and emphasizing the need for demonstration of a use-case analysis for its adoption by the industry. This research presents a digital twin solution to automate the monitoring and controlling of equivalent carbon dioxide (eCO_2) emissions from existing assets through the integration of IoT, BIM, and artificial intelligence across a comprehensive solution, further validating its workability through a real-life use case analysis. The study revealed the significance of BIM and IoT, as essential components of a digital twin to visualise critical spatial information for enhanced facility management specifically for monitoring of indoor air quality of spaces, while also coalescing an AI-supported system to predict carbon emissions from the collected data through integration of machine learning features across the digital twin. The output of the entire solution is displayed as an interactive dashboard for observing trends and patterns, enabling stakeholders to implement effective data-driven retrofitting strategies. This research is a fundamental initiation for implementing digital twins to monitor emissions from existing assets, a step towards achieving the net zero targets.

Keywords: Digital Twin, BIM, Internet of things (IoT), Artificial Intelligence (AI), Big Data (BD), Net zero, Sustainability, Carbon emission

1 Introduction

According to Global Status Report (2017), the built environment is responsible for over 40% of the world's total greenhouse gas (GHG) emissions, with 28% attributed to operation of assets [1]. In context to this, the UK housing sector consumes 25% of the total energy for heating and hot water, further responsible for 16 % of GHG emissions, impacting the climate significantly [2]. The UK net zero strategy is aimed at achieving energy efficiency of housing assets, ensuring utilization of less energy to heat spaces through low carbon technologies by 2035 [3].

The Climate Change Committe (CCC) [4] report on UK housing sector emphasized the need for decarbonization of existing housing stock to achieve net zero ambitions, providing recommendations for policy improvement. According to a research by Dowson, Poole, Harrison and Susman [5], over 80% of the currently inhabited houses will still be in use in 2050, making retrofitting of existing assets a significant challenge for the industry. The most relevant metrics in driving the retrofit works for decarbonization are Energy Performance Certificates (EPC) and Standard Assessment Procedure (SAP) [6], as also identified in another report by the CCC [7] focussing specifically on reducing emissions in Northern Ireland.

Besides implementing retrofit on existing housing stock based on policy objectives, there is a need to monitor the actual performance of assets with real-time robust data, underpinning a substantial gap in as-designed and as-built assets, in accordance with a study by Kelly, Crawford-Brown and Pollitt [8], allowing better implementation of above strategies.

The advancements of industry in Building Information Modelling (BIM) and Digital twin present an opportunity to plug this gap by providing an insight into the utilization of space with real-time information. BIM provides a process for consistent sharing of information for improved collaboration and management across the project's life cycle, encompassing the design, construction and operational phases [9, 10]. However, BIM is limited to provision of static data for the built environment for optimized building performance and energy efficiency, but information cannot be automatically updated in real-time across models and requires additional input [11].

The fourth industrial revolution has been significantly influenced by the emergence of technological enhancements such as Internet of Things (IoT) [12, 13], that allow information exchange of real-time captured data across different platforms and further making possible the integration of this environmental data with BIM Models, leading to the next frontier development in Architecture, Engineering, Construction, and Facility Management (AEC-FM) industry, i.e., Digital twin [14, 15].

Moreover, the proliferation of IoT devices has led to generation of large amount of unstructured Big data, requiring integration with Artificial Intelligence (AI) techniques such as machine learning algorithms, to provide valuable insights and enable informed decision making [16]. A study by Feroz, Zo and Chiravuri [17] identified several of its potentials in built environment, relating to monitoring, and controlling air pollution, optimization of water resources and for efficient energy management of assets, while predicting their impact on climate change, which can then be further integrated with BIM Model.

Digital twin is still nascent to the AEC-FM Industry in comparison to aerospace and manufacturing, with technology being one of its primary challenges for adoption. The term Digital twin was coined by Grieves [18], who acknowledged it as a "virtual replica of the physical asset" interconnecting the two entities with bidirectional flow of real-time information, improvising decision making based on its ability to monitor, compare, predict and conduct what-if analyses. As per a report by Verdantix [19], the digital twin market is estimated to reach a total of $27.6 billion in a period of 10 years, owing to the widespread adoption emerging digital technologies of the industry 4.0, where asset management is expected to gather extreme pace during this period.

The construction industry needs to advance towards utilization of digital technologies for achieving Net zero targets by 2050. Structured and organized data is an essential element of the current industrial transformation which needs to be integrated into the overall infrastructure [20]. The study's motivation stems from the European Commission's action plan, which calls for accelerating investments in digital technologies like smart IoT gadgets and metres, 5G connectivity, Cloud-edge computing, and digital twins of energy systems in order to create a climate-neutral and resource-efficient society, while ensuring beneficial transition for a comfortable and healthy indoor environment.

There is also a rising concern that energy-efficient design strategies such as increasing air tightness measures for the building envelope negatively impacts the quality of the indoor

environment [21, 22]. Indoor air pollutants from materials, appliances, cooking, other activities, or due to inadequate ventilation, have a profound impact on climate change as well as pose further risks to human health beyond those in outdoor air [23]. This requires monitoring of Total Volatile Organic Compounds (TVOCs) and equivalent carbon dioxide (eCO_2) from spaces, as this data can allow to detect problems and track performance of ventilation systems [24], alerting users of their behaviour which can also be inter-linked to energy consumption patterns, a key for achieving net zero buildings as well as healthy indoor environment [25].

Despite significant efforts by researchers in recent years, recognizing the potentials for utilizing digital twin for monitoring and optimization of building performance in real-time, providing interactive feedback for better decision making [15], further enabling sustainability assessment of assets [14, 26, 27], there is a need for a fundamental methodology for transitioning of existing assets to digital twin in the built environment, providing a path for its implementation to industry professionals.

This research is therefore an initiation towards application of digital twin, to monitor indoor emissions from operational assets, through utilization of the most recent technological advancements.

The high-level architecture developed for the digital twin solution in this study is also focussed towards combining data and intelligence through integration of IoT, BIM, AI and machine learning in a single platform, offering a user-friendly interface for monitoring real-time indoor air quality as well as providing an automated process for analysing the recorded time-series data based on feature engineering and SGD technique to forecast future emissions from spaces.

This work is a response to recent recommendations by authors emphasising the need for practical digital twin implementation [28-30], for industry adoption and collaboration, illustrating its implementation on a small-scale housing, by installing IoT sensors measuring temperature, humidity, eCO_2 and TVOC from occupied space, selected based on their substantiated linear dependency [31, 32].

The study employs metal-oxide gas sensor, SGP 30, including multiple sensing elements on one chip, with resistance to contamination from other gases, providing long-term stability for reliable results with quicker response times [33], integrated with AHT20 sensor and ESP32 microcontroller comprising the IoT device component system.

Besides, this research is an initial step towards monitoring of 2050 targets of reduced carbon emissions for housing in the built environment, by implementing digital twin to ensure responsive data-driven retrofitting strategies based on live information feeds for designing efficient ventilation and air purification systems which are a key to reducing pollution and enhancing healthy indoor conditions of spaces, to achieve net zero buildings.

The results validate the Digital twin's capability to display the real-time behaviour of the physical space across the 3D BIM Model, while also predicting the future emissions from space, displaying its similarity in trend as compared to actual emission and indicating poor air quality in bedroom followed by kitchen, necessitating an effective ventilation strategy for these spaces during evening hours.

The articulation of the digital twin artefact through the integration of current technological advancements and its demonstration on existing housing asset, in this research make important contributions to the body of knowledge, presenting their potentials and practical implications to accelerate its application in the domain of facility management.

The research has been organized in 6 main sections, following the introduction in Section 1, the conceptual background is outlined in Section 2, further summarizing the research methods in Section 3. The comprehensive digital twin system for carbon emissions is presented in Section 4 and Section 5 demonstrates its workability through implementing it on an illustrative study. The discussion, significance and limitation are presented in Section 6, finally highlighting the research conclusion in Section 7.

2 Conceptual background

2.1 Application of IoT in the built environment

A study by Begić and Galić [34] concluded that IoT, BIM, Cloud computing and Big Data analytics are the most significant drivers of construction 4.0 across all project phases from

design to construction and operation. IoT holds enormous promise for the development of innovative new applications across a wide range of industries, categorized across 3 major domains: social, environmental, and economical, [35, 36], interconnecting various aspects for improvising the quality of everyday life.

Studies by Kourgiozou, Commin, Dowson, Rovas and Mumovic [37] and AlFaris, Juaidi and Manzano-Agugliaro [38] also revealed IoT as an enabling technology for achieving targets of Net zero buildings, where smart energy systems and smart grid form an important aspect, focussed on improving the efficiency and rationalizing energy consumption [39, 40].

Further developments to leverage the massive data collected from IoT devices can be seen through the implementation of AI techniques to forecast energy consumption, providing insights to users and most importantly reducing its cost implications [41, 42].

Hence, it is evident that IoT plays an important role for achieving sustainability and climate response in the built environment. Prevalent use cases identified include use of IoT sensors, artificial intelligence and cloud computing for efficient building operation summarized in the following Table 1.

A few use-case analysis and frameworks have also demonstrated utilization of BIM for smart building and smart city management for better coordination and control of IoT devices, providing a spatial context to the functioning of the system [52], hence indicating its importance across the operational phase of a project, which has been highlighted in the following section.

Table 1. Review of Application cases of IoT in AEC-FM industry

	Author / Year	Type	Application	Phase
Social + Economical + Environmental	Lin and Cheung [43]		Supply chain management and logistics safety management of workers on site	Construction
	Scott, Bernheim Brush, Krumm, Meyers, Hazas, Hodges and Villar [44]	Residential (Smart home)	Occupancy sensing and Optimization of indoor comfort Automated heating / Cooling	Operation
	Feldmeier and Paradiso [45]	Education Building (University)		

	Al Dakheel, Del Pero, Aste and Leonforte [46] Jia, Komeily, Wang and Srinivasan [47]	Residential (Smart home)	Smart Lighting	Operation
	Viswanath, Yuen, Tushar, Li, Wen, Hu, Chen and Liu [40]	Residential (Smart home)	Maximizing energy consumption	Operation
	[41, 42, 48]	Smart Building	Energy consumption forecasting	Operation
	[35, 49-51]	Smart City	Waste Management Intelligent transportation Environmental Monitoring Public safety and emergency management	Operation

2.2 *Integration of BIM and IoT for asset management*

BIM is a concentric approach to achieving the construction 4.0 aspects, outlined in studies carried out by Maskuriy, Selamat, Ali, Maresova and Krejcar [13] and Zabidin, Belayutham and Ibrahim [53], aggregating vital digital information about a project in a central database and its integration with the various Industry 4.0 technologies has been at the forefront across the built environment. Although BIM can only provide static data of a project and cannot update live information in models instantaneously, however its integration with IoT presents a cyber-physical system that can be used to monitor asset performance in real-time, allowing visualization of critical issues as they may arise in the physical environment across the BIM model [11, 53].

A study by Desogus et al. (2021) considered an easy and straightforward approach for displaying temperature and humidity across the shared parameter field in the Revit BIM Model for existing buildings, by utilization of open-source visual programming Application Dynamo to link the stored sensor data, although limiting it to only 2D visualization of the data.

BIM and IoT integration are most dominantly seen in construction operations monitoring and facility management, accounting for approximately 67% and 22% as per a mapping review conducted by [54], similar to a study by Tang, Shelden, Eastman, Pishdad-Bozorgi and Gao [11] that identified its emerging areas of application and summarized its integration methods, where BIM tools' APIs and relational database has been the most widely adopted as it requires less programming expertise.

Sava, Pluteanu, Tanasiev, Patrascu and Necula [55] presented a case study for using BIM information in Revit for integration with IoT system platform through an external API (Application Programming Interface), particularly for visualization of data from sensors in the model, that focussed on energy efficiency aspects of Net Zero Energy Buildings (NZEB) smart homes.

Transparent information sharing and visualization across stakeholders plays a crucial role in the built environment, enabling sustainability management, which can be achieved through integration of BIM-IoT as determined by Malagnino, Montanaro, Lazoi, Sergi, Corallo and Patrono [20], potentially reducing environmental impact of a project across its whole lifecycle.

BIM and IoT integration promise to transition the construction industry from product to service oriented [56], enhancing data transparency and eliminating the silos for better facility management, nonetheless, it poses technological and interoperability challenges for its utilization, requiring further exploration and research. Malagnino, Montanaro, Lazoi, Sergi, Corallo and Patrono [20] also argued that absence of source code solutions is one of the major deficiencies, limiting its innovative integration.

There have also been limited studies for integration of BIM, IoT, Big Data and AI combined within a single platform, which can enable an enhanced intelligence for the overall system [15, 57].

Artificial Intelligence techniques such as machine learning algorithms, allow to provide valuable insights by analysing Big data, enabling informed decision making [16]. Machine Learning (ML) utilizes structured data with human experts to extract and determine feature sets for the input data, enabling the algorithm to learn from behaviour of the observed and provide its results, emphasizing its prediction accuracy rather than model accuracy [58].

A study by [59] utilized machine learning algorithms, Artificial Neural Network (ANN) and Support Vector Machines (SVM), for predictive maintenance of Mechanical, Electrical and Plumbing (MEP) components, integrating the BIM models, FM systems and IoT network in the information layer, further validating its feasibility through an illustrative example, however the process to train a new model was not automated and required further expertise from facility managers.

Subsequently, several researchers highlighted and concluded the integration between BIM and IoT as a basis for emergence of digital twins in the built environment [9, 14, 15], which has been discussed in the subsequent sections.

2.3 Current State of Digital twin application to achieve Net zero in built environment

The implementation of digital twin was first pioneered by NASA in the aerospace industry, for monitoring, simulating, and analysing the performance of a shuttle to solve any probable issues that it may encounter in space [60], which was then rapidly expended across the manufacturing industry [61].

The conception of a Digital Twin in AEC-FM industry comprises of three major integrating components: Wireless Sensor Network (WSN), data analytics, and 3D model (extracted from BIM) of the physical asset (building, bridge, etc.), facilitating real-time data visualization, comfort enhancement, data-driven decision making as well as improvising building efficiency [62], consequently emphasizing its utilization for existing buildings (retrofitting with sensors) and operational phase of assets.

Sharma, de Sousa Jabbour, Jain and Shishodia [63] presented an overall framework mapping the role of digital technologies enabling achievement of the European Green Deal (EGD) policies and further highlighted the importance of digital twins for design, construction, as well as operation of energy and resource efficient buildings, similar to a study by Çetin, De Wolf and Bocken [64] that was centred towards a circular built environment.

Zaballos, Briones, Massa, Centelles and Caballero [27] provided a framework for integration of BIM model and IoT for sustainability monitoring of asset in use phase, utilizing emotion detection as a smart learning process for comfort monitoring, while research by Tagliabue, Cecconi, Maltese, Rinaldi, Ciribini and Flammini [14], focussed on indoor air quality and energy consumption as well as enabling a decision making mechanism for replacement of materials based on EPD for its maintenance, considering the whole building's life cycle.

In contrast to this, Zhao, Zhang, Wang and Wang [26] presented NZEBs retrofitting proposals for an existing building through laser scanning of asset and further simulating its energy consumption data through DesignBuilder software, lacking utilization of real-time data.

At a macro scale, Park and Yang [65], developed a prototype for GIS-enabled digital twin system for evaluation of carbon emissions, through combination of diverse data sources (Electricity, city gas, household waste, and vehicle) and predicting spatial trends for hot spots, however requiring integration of real-time information.

The current research in the domain of Digital twins and sustainability, indicates an immense focus on energy consumption and its management for operational assets, being one of the key elements for achieving Net zero in the built environment, however, there is a need for an overall integrated framework to monitor, report and control carbon emissions from existing assets, which can subsequently be reduced by implementing retrofitting schemes or offsetting renewable energy generation on-site and off-site.

2.4 Correlation of Indoor Air Quality, TVOC, eCO_2 measurements and Net zero

Indoor air quality (IAQ) is mostly attributed as the interior climate of a building, highly variable dependent on (i) humidity and temperature, (ii) presence of contaminants from gas cookers, stoves, fireplaces, as well as furnishing materials, (iii) building characteristics including infiltration and ventilation rates, and (iv) occupant activities, such as opening of windows, cooking, smoking, or use of consumer products and extractor fans [66].

Poor IAQ is related to sick-building syndrome (SBS), due to lack of adequate outdoor air ventilation, improper exhaust, ventilation of odors, chemicals, or fumes. The major contributors to SBS are presence of Volatile Organic Compounds (VOC) such as benzene, ethylene, formaldehyde etc., diverse set of hazardous organic chemicals that participate in atmospheric photochemical reactions and CO_2 pollutants primarily from occupant's respiration [67, 68].

TVOCs are used as a measure of the total volume of indoor VOC concentrations [69] and acute exposure to these can have a significant negative effect on health of residents as well as impact their cognitive abilities like strategic thinking and decision making [70].

Furthermore, as a consequence of climate change, the impact of indoor sources as well as pollutant levels on air quality in residences may become more prominent due to changes in occupant behaviour and activities (e.g. opening of windows in summer) [22], further also contributing to the GHG emissions from a space. A study by Militello-Hourigan and Miller [71], examined the impacts of cooking activities on indoor air quality of passive houses in Colorado, demonstrating inefficiency of the mechanical ventilation features, contradicting the design assumptions, as well as indicating exceeding CO_2 levels based on ASHRAE comfort limits, leading to recommend increased ventilation in bedroom and direct exhaust range hoods in kitchen.

Similarly, a Building Performance Evaluation study provided insights of five Passive House dwellings located in Scotland, identifying issues with thermal comfort, Mechanical Ventilation with Heat Recovery (MVHR) system performance requiring recommissioning, increased CO_2 concentrations impacting IAQ, and occupant handover for better maintenance [72].

Hence indicating the necessity to monitor the TVOC and CO_2 emissions from space to ensure the as-built performance of assets and their ventilation systems, informing users of their behaviours which can also be linked to energy consumption patterns, thus a crucial factor in realizing net zero buildings and a healthy indoor environment [25].

Moreover, the rapid growth in development of IoT devices as well as low-cost hardware such as metal oxide gas sensors that provide good precision but a low accuracy and their further interface with cloud-based applications provide an opportunity to monitor Indoor environment quality, proposing solutions that can be adopted by practitioners [73].

2.5 Research motivation and Gap

Industry and academia highlight various potentials of digital twin in the built environment providing an in depth understanding of its theoretical framework, however, there have been limited efforts for addressing the fundamental implications of its implementation across a use-case analysis [29, 30], recommending its need by various researchers, summarized in the following Table 2. To advance the adoption of digital twin, there is a need to validate and evaluate the technical aspects of its implementation, as well as resolve interoperability challenges for integration of technologies such as BIM and IoT [74].

The Centre for Digital Built Britain in UK has been advocating towards the development of a national digital twin, aggregating various individual assets and infrastructure built for various purposes, setting out a fundamental standard. However, there has been limited research carried out for mapping of the housing sector, a key segment of the existing assets in the UK, requiring digitization at a large scale, for improving its maintenance and management [75].

A further study is required to address certain research gaps for (1) exploring the emerging digital platforms as well as technologies required for development of an integrated Digital twin instance framework and (2) validation of a simple digital twin solution for industry adoption to develop decarbonization pathways to achieve net zero targets.

Thus, this study bases its motivation from the needs of industry practitioners and other researchers to develop a digital twin solution for existing housing assets to autonomously visualize the environmental parameters of the room across the BIM Model and implement data analytics through ML-based algorithm for prediction of eCO_2 concentrations in a room.

Table 2. A review of literature on digital twin summarizing potential recommendations for future research

S.No.	Author (Year)	Methodology	Focus of study	Limitations/ Recommendations
1.	Davila Delgado and Oyedele [29]	Literature review	Comparison of DT, BIM and CPS in built environment and manufacturing sector	Integration of BIM with digital technologies to achieve DT is not explored
2.	Deng, Menassa and Kamat [9]	Literature review Framework for an advanced Digital Twin for building management	BIM and Digital technologies for developing DT	The research recommended a future focus on industry application
3.	Lu, Xie, Parlikad, Schooling and Konstantinou [15]	Literature review Framework for future development of smart asset management	BIM, DT, Standards, operation, and maintenance of assets	Use case for validation of framework is required for practical application
4.	Camposano, Smolander and Ruippo [74]	Literature review, Qualitative Semi structured interviews	BIM and DT Focussed on Finland AECFM Industry	Design and implementation of technical aspects suggested for further evaluation
5.	Shahzad, Shafiq, Douglas and Kassem [28]	Literature review Semi-Structured Interviews	BIM and Digital technologies for advancement of DT Applications, benefits, and challenges of DT	Requires further clarification for the relationship between DT and other digital technologies with practical challenges based on industry implementation
6.	Khajavi, Motlagh, Jaribion, Werner and Holmström [62]	Literature Review Experimentation using a testbed	Sensor network for developing DT for real-time data collection (light, temperature, relative humidity)	Limited to collection and interpretation of real-time data requiring further integration with virtual BIM Model

3 Methodology

Design Science Research (DSR) is a pragmatic study [76], that focusses on orchestrating a platform or IT artefact for a given context utilizing digital technologies, potentially contributing to digital innovation as suggested by Horita EA, Rocha, Souza and Gonzales [77].

It has been frequently used in information technology and information systems [78], correlating to this study's motivation, i.e., industry 4.0 technological advancements in AEC-FM industry such as IoT, BIM and Digital twin for monitoring carbon emissions, focusing towards achievement of Net zero in the built environment.

Figure 1, provides an overview of the DSR process implemented in this research to achieve its aim and objectives, explained in the subsequent sections.

DSR, however starts with the **awareness of a problem** for which a solution is suggested and then designed in the later stages [79].

A viable problem statement identifies the importance of the study, which drives the study, its aim and objectives, research questions, literature review, methodology, results and conclusions [80], serving as a starting point for the study further unifying the elements of research, leading to a significant contribution as also indicated by Leedy and Ormrod [81].

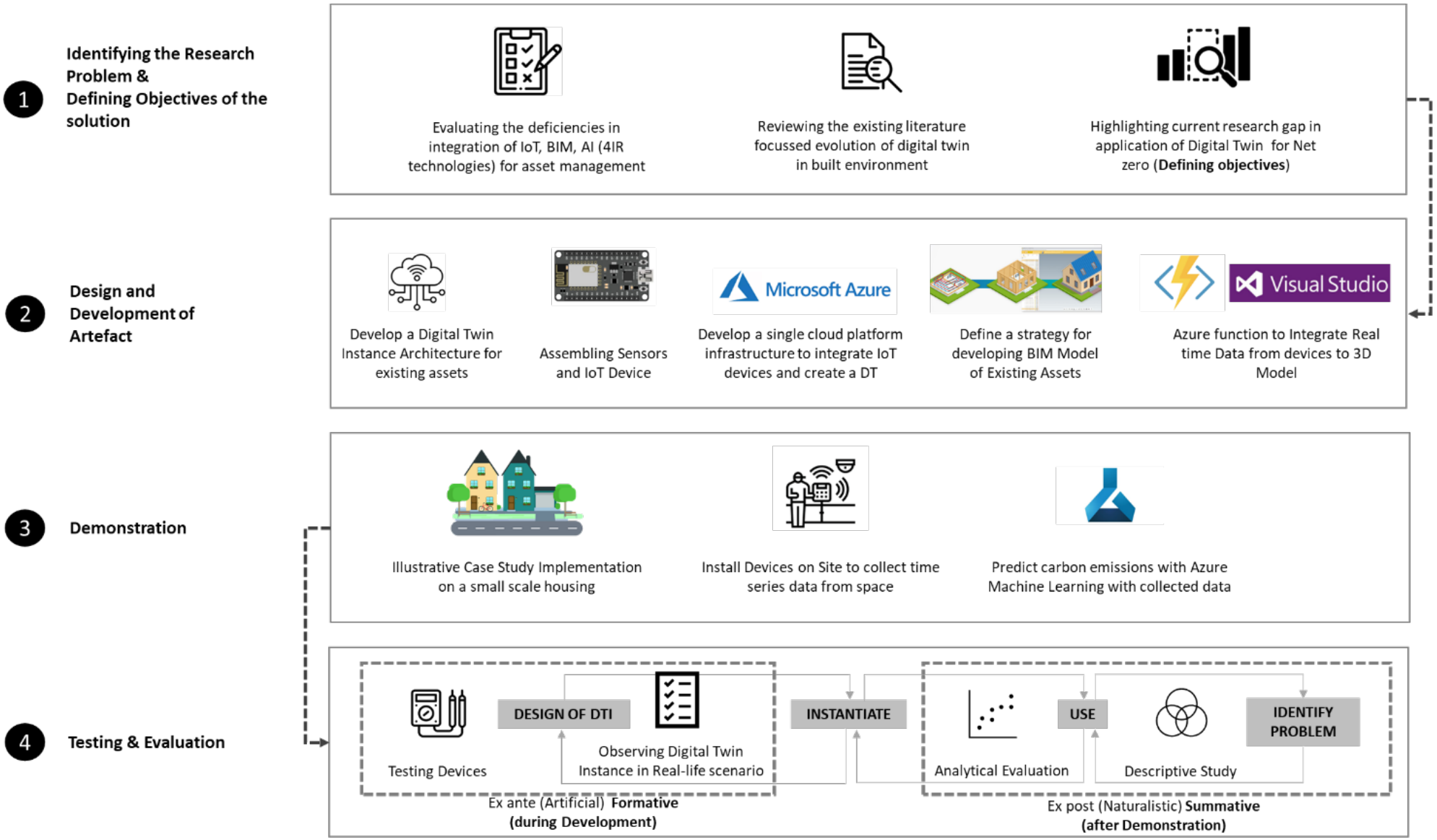


Figure 1. Design Science Research Methodology

3.1 Tools and Techniques

The proposed solution addresses the current technological gap of utilizing digital twin for net zero to monitor and predict carbon emissions in real time, by integrating IoT, 3D BIM Model and AI for analytics in one platform system, based on the identified objectives of the solution development to include "theory of knowledge" as part of the "rigor cycle" [76, 82], improvising the performance of the new artefact proposed.

The following tools are used to develop the infrastructure and test its viability on a residential project:

1. Sensors such as AHT20 as well as SGP30 are assembled along with the ESP 32 development board to create the low-cost IoT device for gathering temperature, relative humidity, TVOC and eCO2 data from space installed. In addition to this Arduino IDE® software is used for establishing connection of IoT Devices to Azure cloud platform to remotely send the telemetry data.
2. A study by Agarwal and Alam [83] suggested the selection of IoT Middleware for a particular use case depends on factors such as its availability, deployment type, cost efficiency, support for the selected hardware utilized, type of communication protocol, storage technologies and type of analytics supported by the platform. Hence, Microsoft® Azure Cloud Platform is utilized including, Azure IoT hub for connection with IoT device, Azure Digital Twin for integrating real time data with 3D BIM Model for visualization, Azure Machine Learning and Power Bi to analyse raw data from devices and predict future emissions, for providing better insights. In comparison to SmartThings, its deployment type is, IaaS (Infrastructure as Service), that enables an integrated suite of services for IoT projects, supported with MQTT protocol, that are adaptable as well as scalable to specific requirements [84]. A comparative review of high-performance computing services of various cloud providers, indicated that although the performance of Azure is not as efficient as Amazon Web Services (AWS), when considering non-Windows server platforms, however its interconnection speed and support available for novice developers makes it more compatible for this research [85].
3. The functions for integrating real time data with digital twin graph and 3D model in Azure Digital twin is facilitated through code development in Microsoft® Visual studio (2022) [85], to trigger the REST API, a serverless solution requiring less infrastructure for developing and maintaining applications. Its performance in terms of execution latency or response times is not as efficient as compared to AWS Lambda and Google cloud functions which can handle massive workloads [86]. Nevertheless, Azure functions' event triggers framework provides the flexibility to collaborate with the different services within the platform and is also more cost effective [87].

4. Python programming language to develop an automated code for predicting emissions in Azure Machine Learning, providing a compute instance to extract features, train and deploy across a continuous adaptive process. Azure Machine Learning is employed as a service because it permits the definition of the data process-flow and the integration with the real-time information stored in the cloud from the devices; furthermore, it is a low-cost and less time-consuming development requiring limited data science knowledge; however, it may not be suitable for complex data structures [88].
5. Lastly, Polycam application was considered for scanning the existing asset, to generate detailed scans of spaces with the LiDAR technology and photogrammetry. Its major limitation is the interoperability with different available phone types and its usability on small scale in terms of ease of access. While Autodesk Revit® was used for creating the 3D BIM Model of asset from point cloud.

3.2 Evaluation of the Proposed Artefact

The design of the artefact follows a formative evaluation at initial stages followed by summative and naturalistic evaluation to include a reasonable rigour for its development, a cyclic process adopted from vom Brocke, Hevner and Maedche [89], ensuring its workability, usability and applicability in a real context, incorporating *ex ante* and *ex post* evaluations [90], as shown in Figure 1.

The first evaluation is centred on the functionality of the IoT devices by **testing** them for failures and defects, prior to installation on site to identify the problems and feasibility. The second evaluation is **observational**, where the Digital twin Instance is implemented on a housing project for an in-depth study of the artefact and solution design. The third evaluation is **analytical** where the trained machine learning model is studied for its dynamic performance quality to predict emissions, to instantiate its robustness. Lastly, a **descriptive evaluation** compares the proposed artefact's effectiveness and efficiency with existing knowledge base.

4 Digital Twin System for carbon emission development

The development of a digital twin solution shall vary from a residential to a commercial building requiring definition of various criteria for each type of project as suggested in a study by Love and Matthews [91], who examined the efficiency gains of implementing digital systems information model that can coalesce to generate business value.

Understanding business and social value of establishing a digital twin instance (DTI) correlate with the BIM process, to define the assessment and need for information management of the solution, necessitating a fundamental hierarchical infrastructure upon which other layers and sensors can be incorporated.

This research focuses on real-time monitoring of indoor air quality and to further predict carbon emissions for operational existing assets, presenting an overview of the **Digital twin instance architecture,** as explained in Figure 2. This provides a starting point for the development of a complex infrastructure which can then enable achievement of Net zero for the housing sector.

4.1 High-Level architecture for the DTCO$_2$ solution

The four major layers of this architecture (Figure 2) comprise of the (i) IoT system component layer for collection, transmission, and storage of data on the cloud platform, (ii) 3D digital Model layer of the existing asset for visualization, (iii) Data and Model Integration layer to replicate the physical assets performance in real time, (iv) the Data Analytics layer for forecasting the future carbon emissions using machine learning algorithms, forming the DTI within a single platform to integrate the various applications.

Microsoft Azure has been considered for the development of this DTI for the research, as it provides a component solution as a single platform service [92, 93] for synchronising visualisation and data analysis, which can be utilized by multiple teams for further processing of information as well as integration of customizable additional services depending on end user requirements.

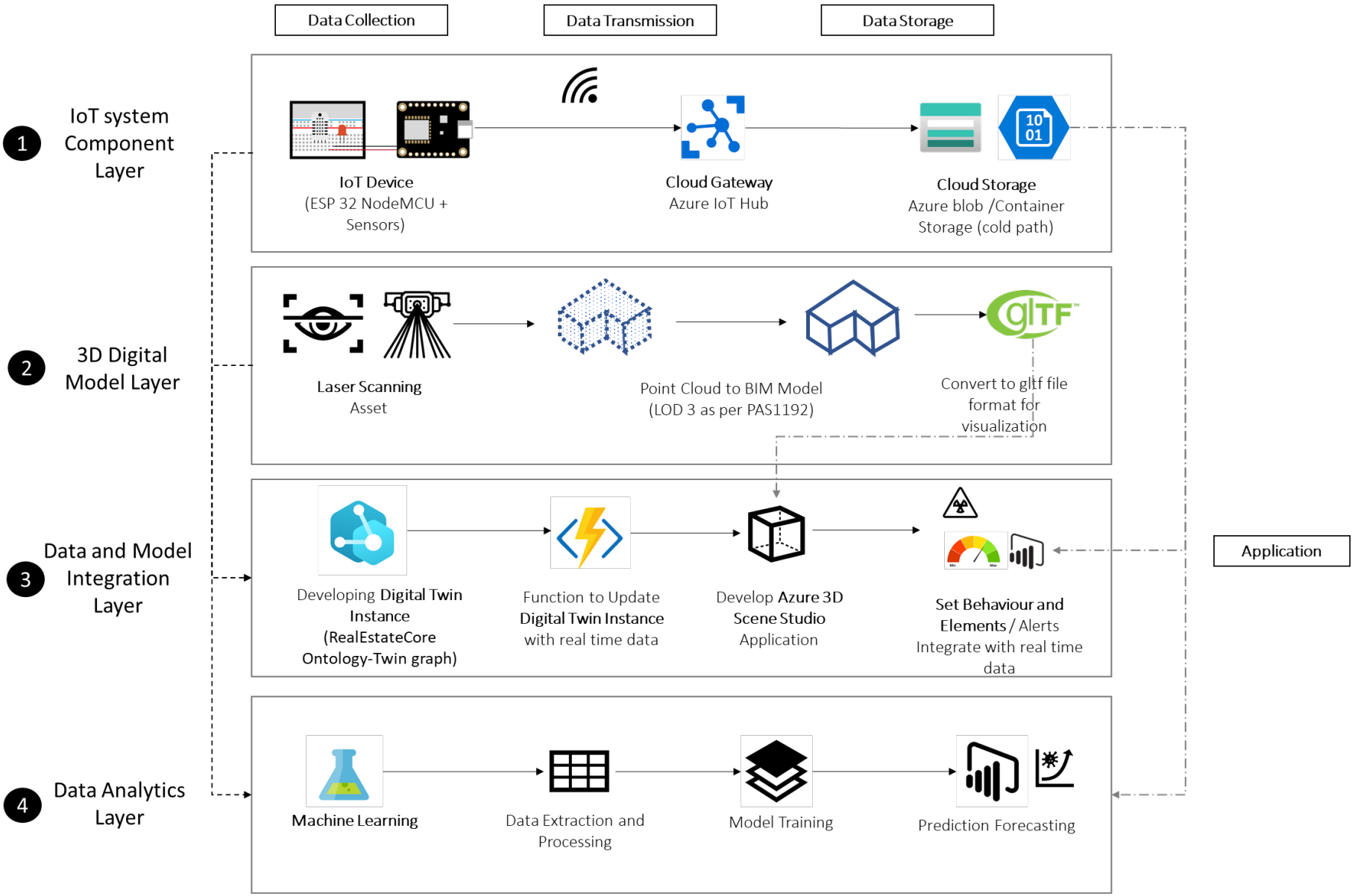


Figure 2. Digital Twin Instance Architecture for monitoring and predicting carbon emissions

4.2 IoT system components (Sensors and Networking)

IoT system component forms the main layer of the DTI architecture, that is responsible for three main functions, i.e., real-time data collection from sensors, its transmission and storage in cloud, correlating to the 3 tier IoT architecture comprising of the perception, network and application layer, usually utilized in the building industry [94].

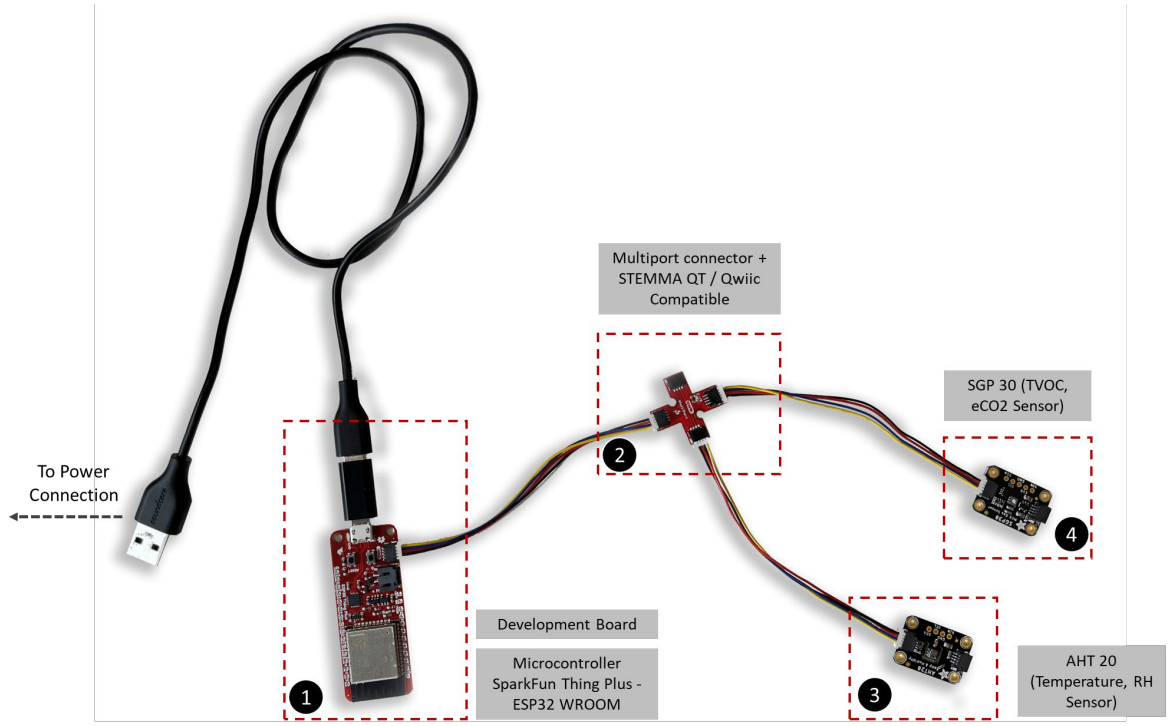
Sensors, actuators, single board computers, smart phones, smart mobile comprise the perception layer, integrating seamlessly with the environment forming wireless sensor networks (WSN) to collect and process data prior to transmitting it to the cloud database [11, 36].

The selection of sensors defines the purpose of this system's solution development, that is monitoring carbon emissions, integrating one data metric required for achieving Net zero. The arrangement of the IoT device components utilized for this study is presented in Figure 3.

I^2C sensors **AHT20** acquires **temperature and relative humidity**, while **Adafruit SGP30 Air Quality Sensor Breakout** measures **TVOC and equivalent CO2 (eCO_2)** of the space connected to an ESP 32 microcontroller, which is more cost effective in comparison to Raspberry Pi and capable of multiple wireless node networking [95]. All the sensors and development boards use STEMMA QT or Qwiic (4-pin JST) connector system, enabling solderless IoT device assembly for quick prototyping including addition of sensors as required [96].

SGP 30 sensor mainly detects various VOC (volatile organic compounds) and H_2 in the air and is programmed to estimate TVOC and eCO_2 concentration measurements, allowing to identify the emissions from a space. The eCO_2 information from a Metal-oxide (MOX) sensor clearly captures more air-pollution events than just a CO_2 measurement, providing information scaled from presence of volatile organic compounds in indoor environments mainly originating from sources of human activities, building materials and appliances, via proprietary algorithms [33].

It must however, be calibrated based on its adjustment algorithm, requiring it to run for at least 12 hours to set and store the baseline values [97].



IoT Device Assembly for one room						
Development Board and Accessories	Sensor	Information	Accuracy	Measuring Range	Range	I2C Address
Microcontroller SparkFun Thing Plus - ESP32 WROOM 	Adafruit AHT20 - Temperature & Humidity Sensor Breakout Board 	**Temperature**	+-1 °C	-40 to 85 °C	Room	0x38
		Relative Humidity	+-3 %	0 to 100 RH%		
Multiport connector + STEMMA QT / Qwiic Compatible 	Adafruit SGP30 Air Quality Sensor- TVOC and eCO2 sensor 	**TVOC**	1-32 ppb	0 ppb to 60000 ppb	Room	0x58
		eCO2	1-31 ppm	400 ppm to 60000 ppm		

Figure 3. IoT Device Assembly and Technical Specifications

Several studies have identified the dependency between TVOC and eCO_2, indicating a linear relationship between their levels, which is further affected by temperature and humidity [98].

The ESP32 board is compatible with **Arduino IDE**, an open-source software that allows testing and reading sensor data by uploading a code (or sketch) [99], relatively used for simple projects, due to its easy configuration and available community support.

The Arduino IDE platform is therefore utilized to develop the software for processing data transmission to cloud from the microcontroller.

4.3 Data Transmission and Storage in IoT cloud platform

Transmission of data to cloud requires to conform to the communication protocol standards of the network layer that is composed of different sub-layers, that utilize network technologies such as WiFi, Bluetooth, 3G, ZigBee, etc [94].

A study by Yassein and Shatnawi [100] determined that MQTT (Messaging Queue Telemetry), an open-source lightweight messaging protocol is most appropriate for IoT applications linking remote embedded sensors and for machine-to-machine communications.

Hence, MQTT is used for cloud protocol adaption in this research as it facilitates event-driven message exchange through Wi-Fi (wireless network) to Azure IoT hub.

The following network diagram and sketch illustrates the process of transmitting the telemetry data to IoT hub, utilizing **Azure SDK for Embedded C IoT Hub Client on Espressif ESP32** through Arduino IDE**,** open-source code for constrained devices.

The device is programmed to send telemetry periodically after 32,500 milliseconds to allow consistent data transmission over a day from one device due to the messaging restriction of 8000 per day based on IoT hub tier selection.

The messages received in the IoT hub can then be further routed to a cold path for long term storage in containers where it records data from each sensor in semi-structured cloud database, allowing it to be retrieved for analytics and insights in future [101], for instance machine learning to provide emission forecasts or comparing monthly data.

The enqueued messages are written to a blob in the container with a batch frequency of 553 seconds and a maximum limit for its chunk size of 500 MB, as illustrated in Figure 4. It is to be noted the file format for encoding has been selected as JSON, which has been also specified as the format for message properties.

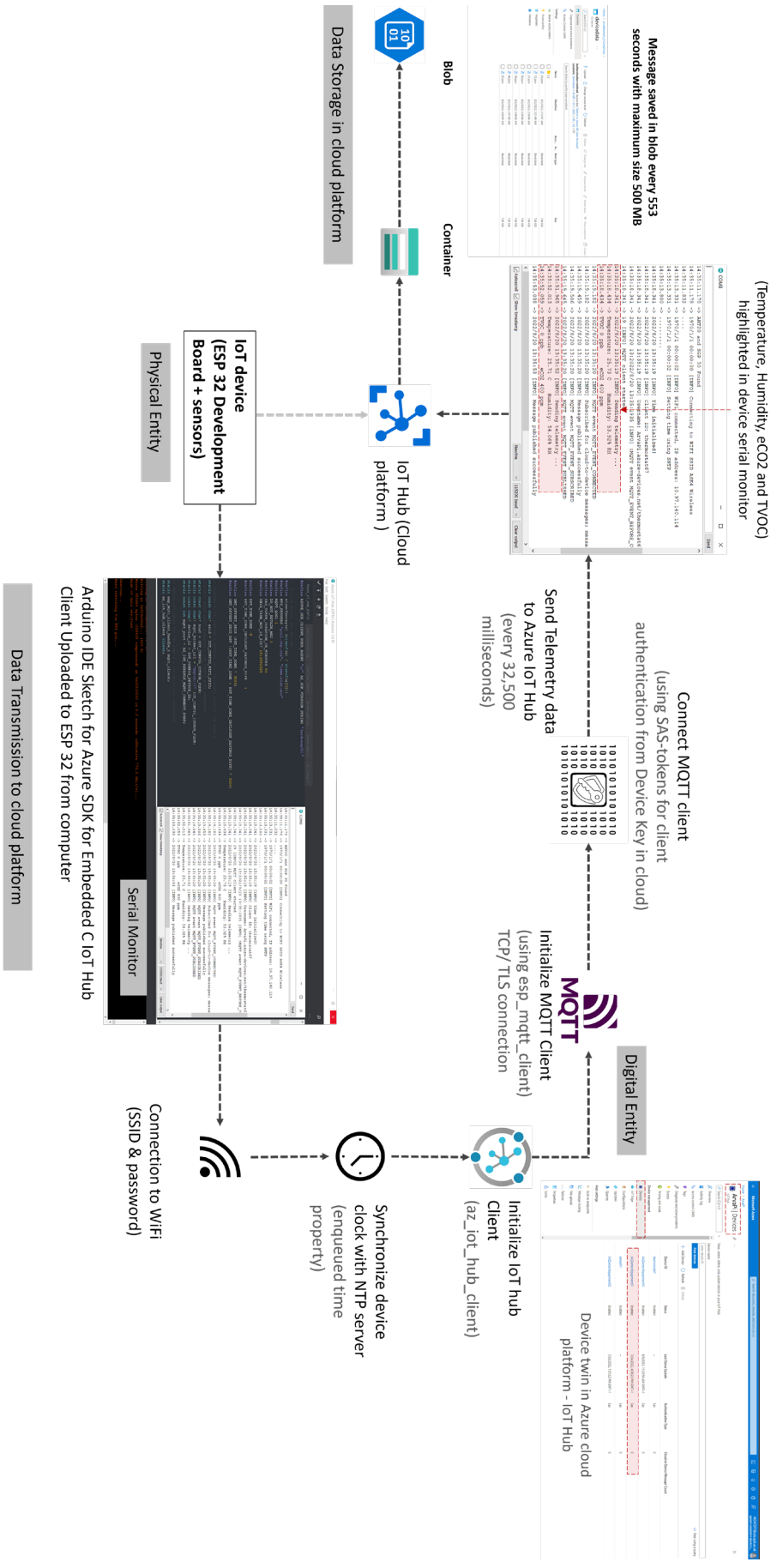


Figure 4. Block Network Diagram illustrating Data Transmission and Storage in Azure cloud platform

4.4 Data and Model Integration Layer

Digital twin enables a seamless transmission of data between the physical and digital world [62], hence this layer forms the core aspect of this architecture, facilitating the monitoring of collected sensor information across the 3D Model in real time.

The Azure Digital twin (ADT) Instance, a platform as a service (PaaS), is utilized to create knowledge graphs replicating the digital model of entire environments, required for interaction of model and data.

This layer includes two main key aspects for working of the DTI, that is defining twins as well as their relations to represent the physical environment and updating of twins with real-time information from physical sensors, as illustrated in Figure 5, and explained in sections below.

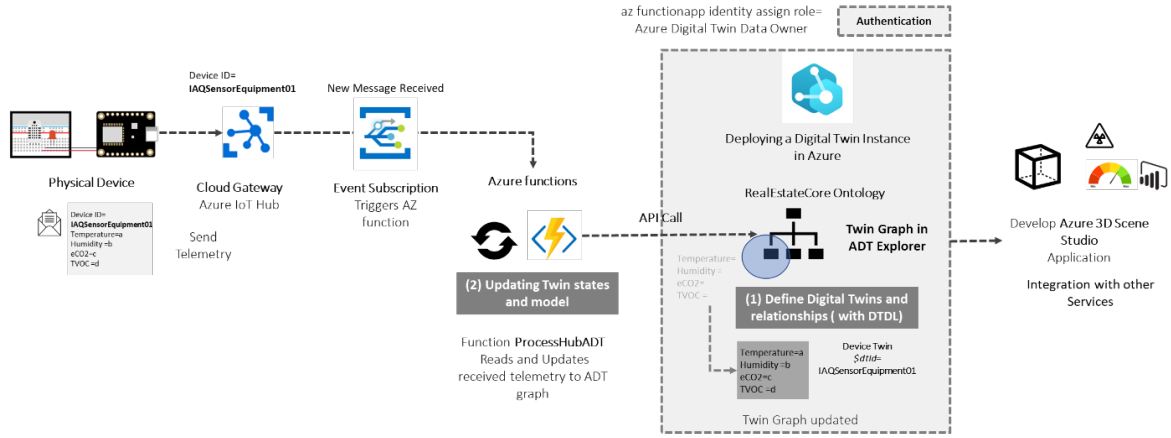
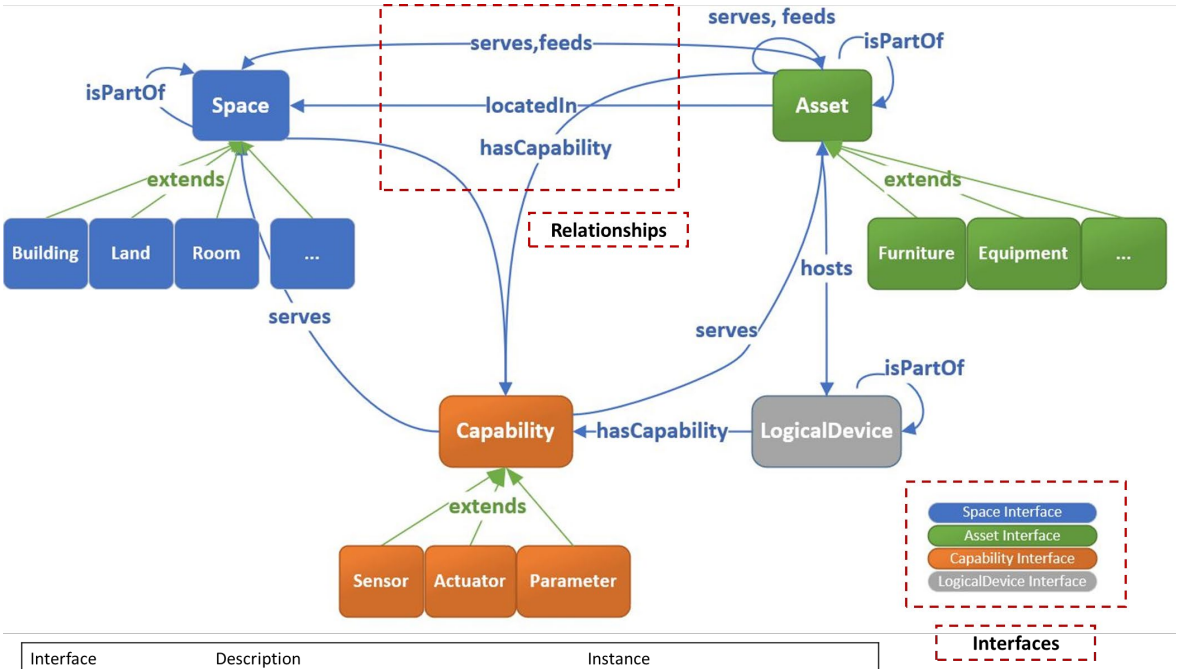


Figure 5. Components of data and model integration layer, process flow of updating Telemetry received from IoT devices to Azure Digital Twin Instance

4.4.1 Define Digital Twins and relationships

Azure digital Twins utilize the Digital Twin Definition Language (DTDL) to model entities, defining the properties and relationships of the physical device in the domain. As part of this research the DTDL-based RealEstateCore ontology has been adopted to model the physical environment and create twin graphs, which can then be integrated with IoT device information, BIM (IFC) and Building Management Systems (BMS). It is an open-source ontology definition based on industry standards (like BRICK Schema) for smart building solutions, meeting the AEC-FM industry requirements [102].

The following graph (Figure 6) indicates a standardized structure of Real EstateCore ontology which can be utilized for any project type and extended according to requirements, constituting of interfaces and relationships.



Interface	Description	Instance
Space	Part of physical world, 3D spatial extent	Site, Building, Level, Room
Asset	Object inside a building (not integral)	Furniture, HVAC Equipment
Logical device	Device or software interacting with Digital twin	Integrated Circuits, virtual Server
Capability	Capacity of an entity to produce or ingest data (Sensors or actuators behaviour)	Temperature, Air Flow

Figure 6. Structure of RealEstateCore ontology

The ontology includes each of these interfaces defined as models using the DTDL language, following the JSON syntax [103] and uploaded to the ADT explorer to create twins and overall graph of the physical environment being studied, a sample is illustrated in Figure 7.

The IAQSensorEquipment contains the structure as explained with Property such as Temperature, which is Semantic Type including value and unit, while the Capability model defines the relationships such as “serves” and “isCapabilityOf” (Figure 7). The device ID representation in IoT hub is used to map the *$dtId* of IAQSensorEquipment twin created in ADT explorer, utilized by the Azure function to integrate data received from physical devices in IoT Hub to the twin’s Property (e.g.: “Temperature”). This is a consistent parameter utilized at device-end for communication and at cloud-end for integration with other services. These twins in ADT explorer graph can then be further linked to the elements of the 3D Model Scene studio to semantically map the mesh, as illustrated in Section 5.3.

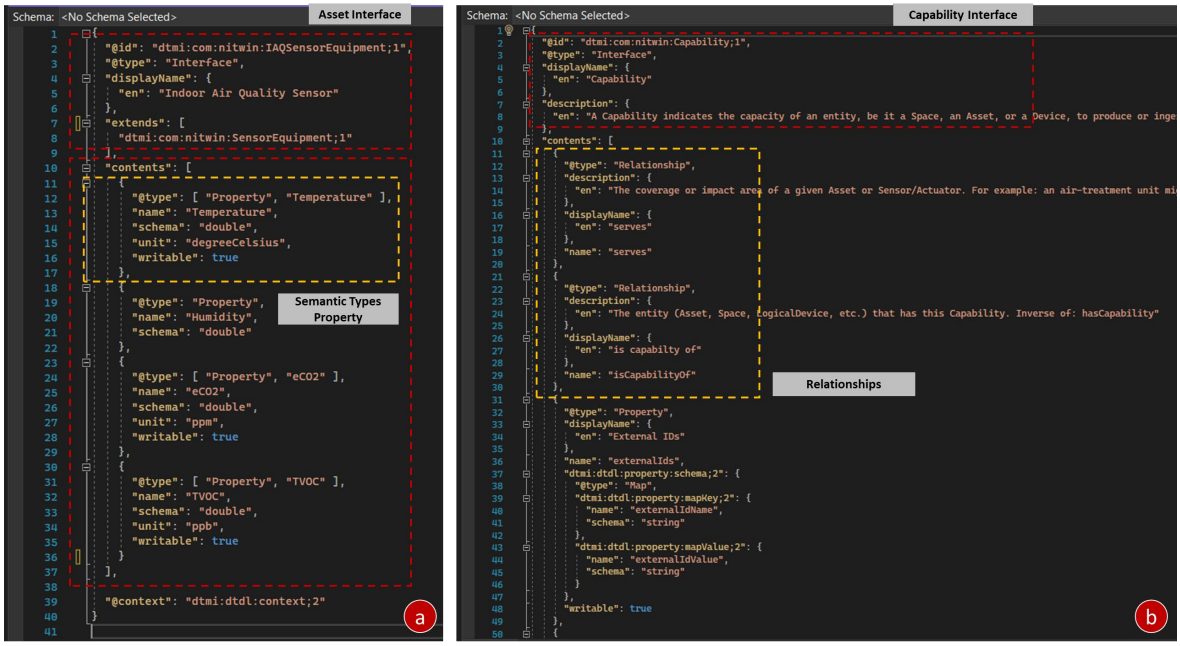


Figure 7. (a) Sample of IAQSensorEquipment Model with Semantic Types Property and Telemetry field, (b) Sample of Capability Model with Relationship field

4.4.2 Updating Twin states and model

To ensure a secure decentralized connection of DT components with the physical device, a serverless RESTAPI is used that is Azure functions allowing to integrate the front-end website (ADT Instance) to the back-end database (IoT Hub), deployed and executed as required, a mediator communicating via HTTP request to create, read, update, and delete records (CRUD) while also maintaining security and control through authentication.

The workflow shown as per Figure 5, highlights the back-end process involved in updating of real-time telemetry data from IoT device to the properties, temperature, humidity, eCO_2 and TVOC across the IAQEquipmentSensor01 twin in ADT explorer graph. The event grid subscription, on receipt of messages in IoT Hub from IoT device, triggers the Azure "ADTSampleFunctionsApp" which is authenticated using the assigned role identity (Azure Digital Twin Data Owner) to read the received information and update it in ADT twin graph via HTTP.

4.5 Data Analytics Layer

The overall objective of the predictive modelling analysis was to leverage the carbon emission data being collected and develop a machine learning model to forecast future emissions, an artificial intelligence technique.

Figure 8 provides a comprehensive machine learning process for prediction of CO_2 emissions from the stored historic data of the rooms.

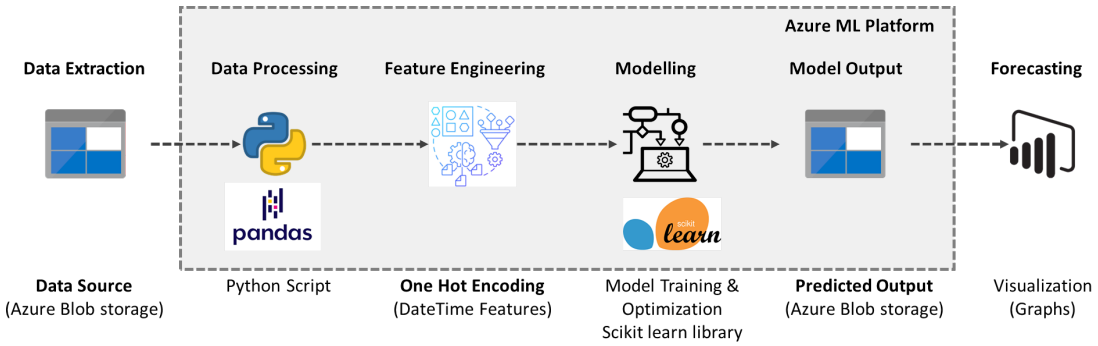


Figure 8. Machine Learning process for prediction of CO_2 emissions

A python script is executed on the Azure ML platform daily to predict CO_2 emissions over the next 24 hours for each room, further allowing to calculate an average for the entire unit (Figure 9). Based on the availability of time series data, new models can be trained every 15 days to ensure its reliability over time. The initial step in the training process involves extracting of sensor readings from Azure Blob Storage container. The processing functions transform the unstructured data stored in JSON format to tabular data suitable for machine learning. As sensor readings are collected within short intervals of 2.5 minutes, hence these CO_2 emissions are grouped together to give an hourly average.

Feature engineering is the next most important aspect, where new features or attributes are created out of existing data. Here, the main focus was on extracting DateTime features that capture the cyclic nature of the hour, day, and day-of-week attributes using sine and cosine transformations. Additionally, a binary flag is used for identifying weekends. To wrap up the feature engineering, one-hot encoding is used to represent the IoT devices in different rooms (Figure 10). Hence the input features of the model include Datetime, Equipment ID (for location) and the average of eCO_2 emissions corresponding to that, which is then utilized for model training and testing.

```python
# creating prediction dataframe
predict_data = {"timestamp":[],"equipment_id":[]}
for equipment in equipments:
    for i in range(24):
        timestamp = pd.to_datetime(current_date) + pd.Timedelta(hours=i)
        predict_data["timestamp"].append(timestamp)
        predict_data["equipment_id"].append(equipment)

predict_df = pd.DataFrame(data=predict_data)
predict_df["timestamp"] = pd.to_datetime(predict_df["timestamp"])

# encoding categorical fields
featurized_df = encode_categorical(data=predict_df,categorical_fields=categorical_fields,train=False)

# extracting datetime features
featurized_df = extract_datetime_features(data=featurized_df)
featurized_df = featurized_df.drop(labels=["timestamp"],axis=1)

# loading the trained model
with open("model.pkl","rb") as f:
    model = pickle.load(f)

# predicting co2 emissions
predicted_emissions = model.predict(featurized_df)

# adding the predicted emissions to output data
predict_df["predicted_emissions"] = predicted_emissions
predict_df["predicted_emissions"] = predict_df["predicted_emissions"].apply(lambda x: math.e**x - 1)

# saving the file to local path
output_data = pd.read_csv("predicted_data.csv",parse_dates=["timestamp"])
output_data = pd.concat(objs=[output_data,predict_df],axis=0)
output_data.to_csv("predicted_data.csv",index=False)

# uploading predictions to blob storage
block_blob_service = BlockBlobService(account_name=account_name_output,account_key=account_key_output)
block_blob_service.create_blob_from_path(container_name="predictions",blob_name="predicted_data.csv",file_path="predicted_data.csv")
```

Creating a dataframe table for representing each equipment with timestamps for the next 24 hours

Extracting Datetime features

Loading the Trained Model

Predicting CO2 Emissions and storing as blob

Figure 9. Python Script to execute the prediction in Azure ML Platform

```python
def extract_datetime_features(data):
    '''
    Function returns binary values based on day type
    '''

    # sine and cosine based on hour
    data["hour"] = data["timestamp"].apply(lambda x: x.hour)
    data["hour_x"] = data["hour"].apply(func=datetime_sine,args=(24,))
    data["hour_y"] = data["hour"].apply(func=datetime_cosine,args=(24,))

    # sine and cosine based on day
    data["day"] = data["timestamp"].apply(lambda x: x.day) - 1
    data["day_x"] = data["day"].apply(func=datetime_sine,args=(31,))
    data["day_y"] = data["day"].apply(func=datetime_cosine,args=(31,))

    # sine and cosine based on day of week
    data["dayofweek"] = data["timestamp"].apply(lambda x: x.dayofweek)
    data["dayofweek_x"] = data["dayofweek"].apply(func=datetime_sine,args=(7,))
    data["dayofweek_y"] = data["dayofweek"].apply(func=datetime_cosine,args=(7,))

    # binary feature is_weekend
    data["is_weekend"] = data["dayofweek"].apply(func=weekend_binary)

    # dropping columns not rquired further
    drop_columns = ["hour","day","dayofweek"]
    data.drop(labels=drop_columns,axis=1,inplace=True)

    return data
```

(a) Feature Engineering to extract datetime features through sine & cosine functions

Original

Equipment
Living Room (IAQSensorEquipment01)
Kitchen (IAQSensorEquipment02)
Bedroom (IAQSensorEquipment03)

One-hot encoded

Living Room	Kitchen	Bedroom
1	0	0
0	1	0
0	0	1

(b) One Hot Encoding

Figure 10. Feature Engineering to create DateTime features from real-time data and one-hot encoding for representation of devices

Before training a machine learning model, the data is split into training and test sets. A time-based split is performed to avoid data leakage, as shown in Figure 11. The data is sorted according to time before splitting into training and testing without shuffling, which is more desirable in this situation so as to train the model on the past values to predict the future emissions. Hence, the time-based split functionality has been utilized in the code, to

divide the data into first 80% for training and the last 20% for testing (Figure 12). To further enhance this the Sklearn TimeSeriesSplit functionality can also be utilized.

Furthermore, a log transformation has been proposed as a means of reducing the skewness of the emissions variable as a result of device malfunctioning which may be attributable to operational environment or connectivity issues. This has been implemented on the target variable because there are large non-negative values recorded for CO2 emissions, across a particular time-period, as this is also desirable for the linear regression algorithm which is used as a way to train the model for achieving efficiency in prediction results.

```
178
179 def split_train_test(data,test_size):
180     '''
181     Function performs time based split into train and test
182     '''
183
184     # sort by timestamp
185     data.sort_values(by="timestamp",inplace=True)
186
187     # log transformation of target variable
188     data["emissions"] = data["emissions"].apply(lambda x: np.log1p(x))
189
190     # splitting into train and test without shuffling
191     X,y = data.drop(labels=["timestamp","emissions"],axis=1), data["emissions"]
192     X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=test_size, shuffle=False)
193
194     return X_train, X_test, y_train, y_test
195
```

Figure 11. Functions used for data processing as part of the Python script

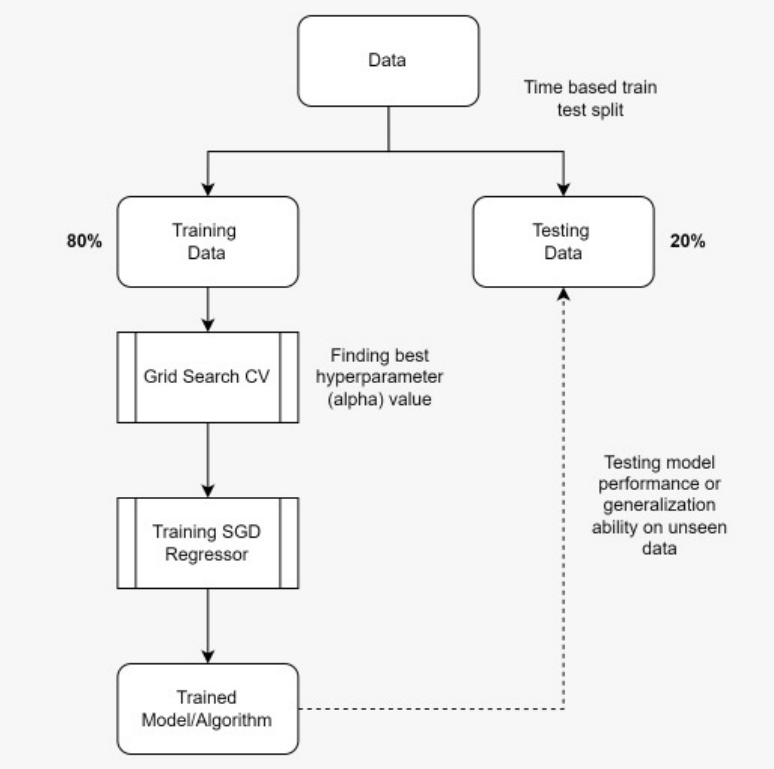
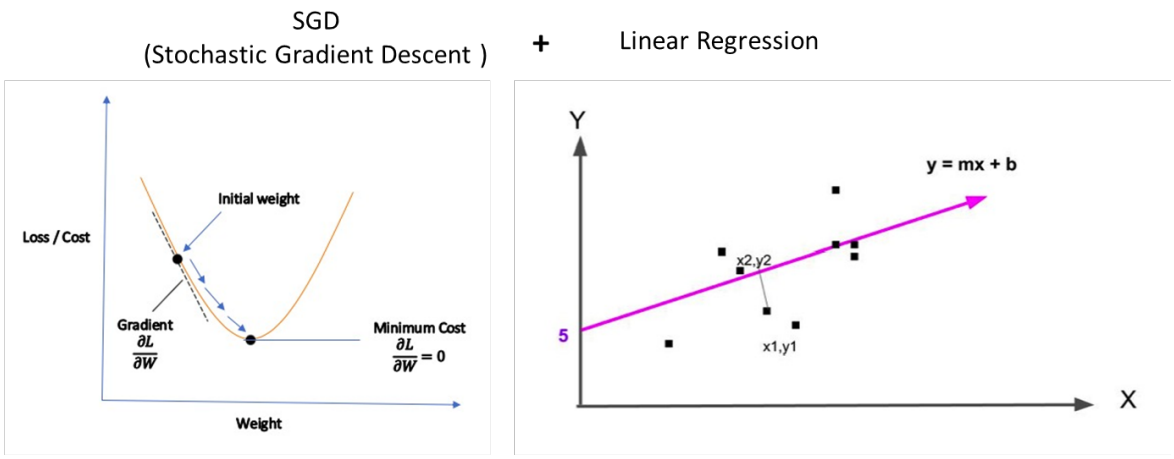


Figure 12. Time based data training and testing

Currently, a single model is used to experiment the Stochastic Gradient Descent (SGD) regressor with L2 regularization. The SGD is an optimization technique used to improve the model's squared error, where the gradient of loss is estimated for each sample at a time and the model is updated as it progresses with a decreasing strength schedule or learning rate.



X= feature or variable
Y = variable to predict (in our case CO2 emission)

Figure 13. Stochastic Gradient Descent (SGD) regressor with L2 regularization algorithm

The SGD regressor avoids overfitting by performing a grid search cross-validation over a range of values for alpha or the L2 regularization rate. Once the best parameters have been identified, the model is saved and trained for future predictions. To evaluate the model, the squared error is computed between the actual and forecasted emissions. The model error is compared against the error of a baseline model (a model that simply outputs the predictions as the average emission for each observation). Once the model has been evaluated, as mentioned earlier it is used to forecast the CO_2 emissions for the next 24 hours. Finally, the predicted emissions are stored in Azure Blob Storage container, which can be visualized later in a dashboard to understand trends and patterns, providing better insights.

5 Testing and Evaluation

5.1 Overview of Case study

An existing dwelling, which is located in Stranmillis Gardens, Belfast, is selected for evaluating the developed solution as shown in Figure 14.

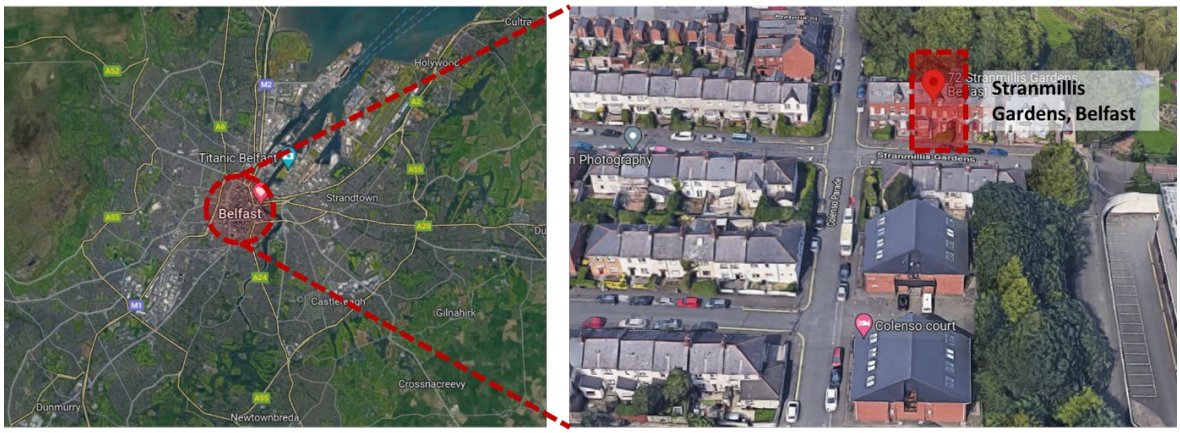


Figure 14. Case Study location map accessed from Google Maps [104]

The solution is implemented only across the Ground floor, one bedroom apartment unit with an approximate area of 48 sqm as displayed in the architectural layout. The apartment selected for the period of this study was occupied by two university students of age ranging from 25-30 years who were also employed in part-time work. The record of number of occupants during this measurement period is critical as the eCO_2 levels correspond directly to the occupancy level of the space, that shall also enable a better understanding of the usage pattern of the household.

5.2 Device Installation and data transmission testing

The location of sensors is dictated by their range to measure the temperature, humidity, eCO_2 and TVOC across a space. The 3 assembled IoT devices comprising of ESP32 development board, AHT20 Sensor and SGP30 Air Quality Sensor are installed in the living room, kitchen, and bedroom to monitor emissions (see Figure 15). These areas are identified as probable sources of emissions for this case study, considering most frequently occupied spaces in a house. The placement of sensors ensures avoidance of encroachment and disturbance from daily activities of residents.

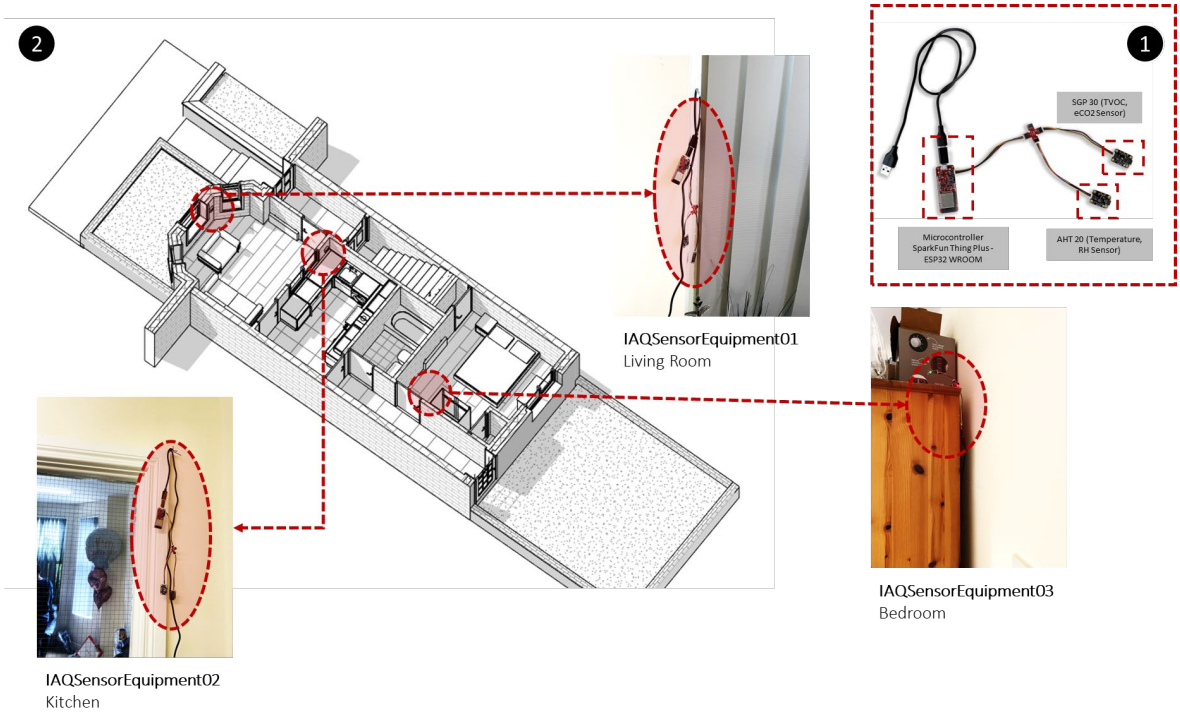


Figure 15. Assembled IoT Device (1) and their placement in Living, Kitchen, and Bedroom (2)

In order to ensure connection of devices to WiFi, the Arduino IDE sketch, as explained in Section 4.3, is modified to include "SSID and password" available for the space and uploaded to the device through USB-COM-port of a computer. The serial monitor in Arduino IDE, provides an information log of successful connection to MQTT as well as IoT Hub while also displaying the temperature, humidity, eCO_2 and TVOC data recorded by device to ensure its working or troubleshooting errors, as shown in Figure 16.

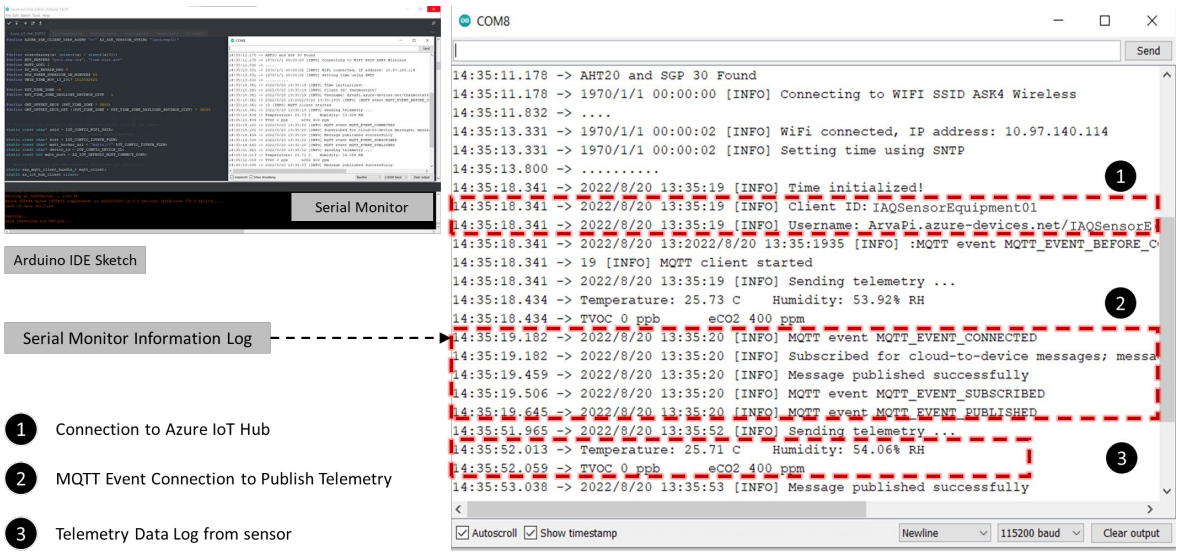


Figure 16. Serial Log monitor for testing of device connection prior to placement in location for IAQEquipmentSensor01 (located in Living Room)

Once tested, the devices are placed in their location, powered through a USB Adapter, and reset to automatically connect to WiFi. The uploaded sketch on ESP 32 microcontroller processes every 2.5 minutes to send telemetry recorded to Azure IoT Hub, which can be seen on its dashboard presented in Figure 17. The received data is further linked to Azure Digital Twin graph for better insights and visualization in 3D Model.



Figure 17. Azure IoT Dashboard for remote monitoring showing device connections and cloud messages received

5.3 3D Asset and Digital Twin Graph

The existing asset's 3D Model was developed to enable visualization of incoming data in the cloud platform, for the proposed DTI solution.

Polycam was utilized for LiDAR scanning of this research development; however, for implementation at macro scale terrestrial laser scanner can be utilized for enhancement. The scanned assets' model is exported to point cloud in .dxf format, which can be imported in software such as Autodesk Revit to develop LOD 200, 3D BIM Model (Figure 18), constituting the architectural elements and spaces, which can be further enhanced by including comprehensive Structural and MEP services for a more robust building information model.



Figure 18. Overlay of imported point cloud in dxf and developed 3D BIM Model

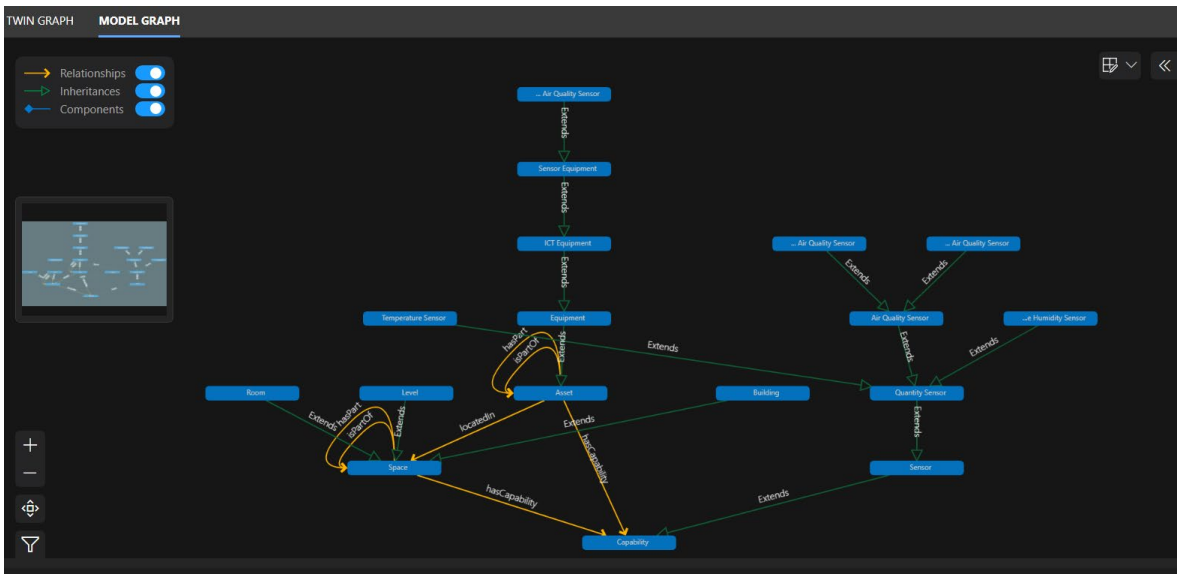


Figure 19. Twin Graph Models, relationships and Components as uploaded from RealEstateCore Ontology to ADT Explorer

The final exported **glTF model** comprising of mesh and textures is uploaded to 3D scene studio for further integration with Digital twin graph in ADT explorer as explained in Section 4.4.1. To develop the asset environment in ADT explorer, the main interface models, Space, Assets, Capabilities and Sensor were uploaded with their sub models, to present the Model Graph as shown in Figure 19, based on the RealEstateCore Ontology industry standards.

The twins IAQSensorEquiment01, IAQSensorEquiment02, and IAQSensorEquiment03, were created in the Twin Graph representing the capabilities of sensors located in living room, kitchen, and bedroom respectively, which are further a part of the ground floor and the building subsequently, as per their relationships, illustrated in Figure 20. A set of meta-data or properties are associated with the twin which can be updated to include additional information.

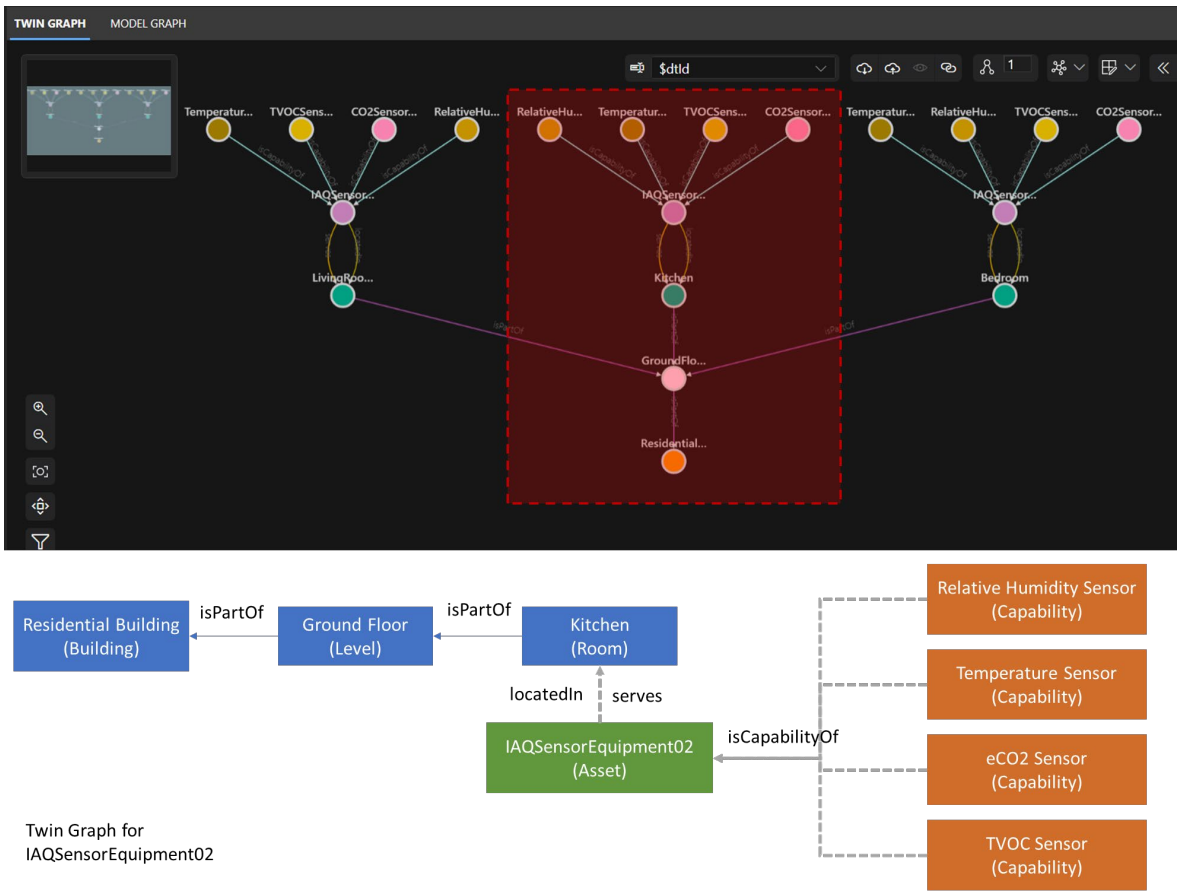
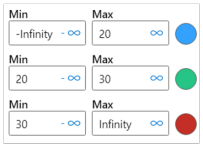
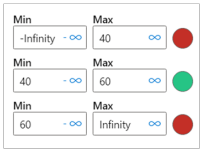
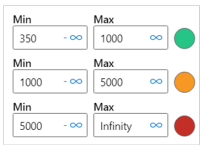


Figure 20. Digital Twin Graph to represent the environment of residential asset being monitored

The following Table 3 shows the Behaviour, Status, and Alert criteria of Elements implemented in 3D Scene Studio, enabling monitoring and visualisation of temperature, humidity, and eCO_2 data received from sensors located in the rooms across the 3D Model.

Table 3. Overview of the behaviours and elements for visualization insights of data in 3D Scene Studio

3D Scene Studio Element Behaviours for Visualization				
Elements	Behaviour	Status	Alert	Widget & Preview
IAQSensorEquipment01 (Living Room) IAQSensorEquipment02 (Kitchen) IAQSensorEquipment03 (Bedroom)	**Temperature**		PrimaryTwin.Temperature > 30  ${PrimaryTwin. $dtId} is displaying very high relative humidity in room	
	Relative Humidity		PrimaryTwin.Humidity >60  ${PrimaryTwin. $dtId} is displaying very high relative humidity in room	
	eCO2		PrimaryTwin.eCO2 >1000  High CO2 emission recorded from ${PrimaryTwin.$dtId}	

As per ASHRAE Standard 55, the acceptable thermal conditions for operational buildings, i.e., the temperature of a space must be within 19 to 29 °C, while a recommended relative humidity of 30 -60%, above which is noted as discomfort levels for occupants.

CO_2 emissions from a space not only impact the environment but also impact its occupants and is an indicator of Indoor Air Quality (IAQ) and Ventilation requirement as per ASHRAE and the same criteria are utilized for visualization.

5.4 Real-time data visualization

The information from IoT devices can be presented in the Twin graph and in the 3D Model in Scene Studio. The live data streamed from devices to IoT hub is linked to the Twin Graph where the properties (temperature, humidity, eCO_2, TVOC) are read and updated based on the Device ID by activating the Web API, “ADTSampleFunctionsApp”.

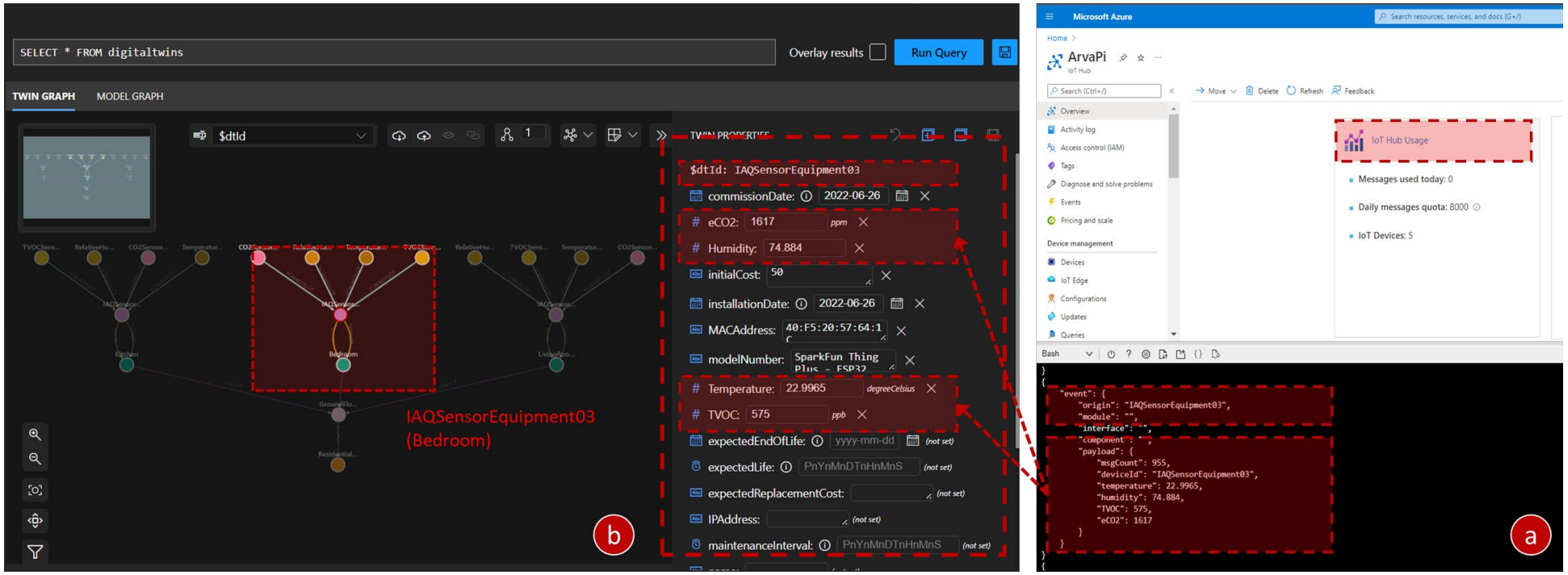


Figure 21. Twin graph indicating the current received values from IoT Device, (a) Message payload received from MQTT to cloud from IAQSensorEquipment03 located in Bedroom, (b) Real-time telemetry data updated across the properties of IAQSensorEquipment03 in twin graph



Figure 22. Interactive 3D Visualization of Asset with real time emissions and indoor air quality monitoring (a) eCO_2 Emission Layer indicating high emissions from Bedroom (shaded in yellow) (more than 1000 ppm), (b) Temperature displayed with the status across the model, (c) Humidity Layer showing high relative humidity (more than 60%) across all rooms

As seen from the above Figure 21, information recorded from sensor in bedroom (IAQSensorEquipment03), is received in the IoT Hub cloud platform through the MQTT broker, which is then further updated across the properties of its twin in the graph.

The 3D Model visualization provides an interactive interface, as shown in Figure 22, where the received data in twin graph, displays spaces, based on status and alerts which have been defined. For instance, in the demonstration, when eCO_2 emissions are recorded beyond the comfortable range, i.e., 1617 ppm (>1000ppm) (Figure 22(a)), an alert is generated and floor plan shaded to yellow from base, emphasizing need for increased ventilation of space.

Hence, this allows end users to ensure indoor air quality while providing critical information about emissions from a space in real-time. Further, this information can be analysed to identify patterns as well as estimate emissions, for simulating retrofitting solutions and their impact, enabling informed decision making by asset managers, for achievement of Net zero strategies.

5.5 Prediction Dashboard

The real time data stored from the three devices, comprising of 8000 messages per day in the cloud platform, is then utilized to predict the eCO_2 emissions on hourly basis through machine learning process explained in Section 4.5. This output is visualized in PowerBi Dashboard providing a comparison between actual and predicted emissions as shown in Figure 23. The whole dataset consists of 2,91,805 records of eCO_2, TVOC, Temperature and Relative Humidity measures. In order to make them comparable, this information was interpolated into a value of each parameter for every 60 minutes. The mean squared error of the last trained model was noted to be 0.6, as compared to the baseline model error, which was 1.049, during the test period, further indicating the improved predictability of eCO_2 emissions. The line charts show the trending of actual eCO_2 emissions and predicted emissions, observing the most emissions from bedroom followed by living room and kitchen respectively, indicating inadequate ventilation of space.

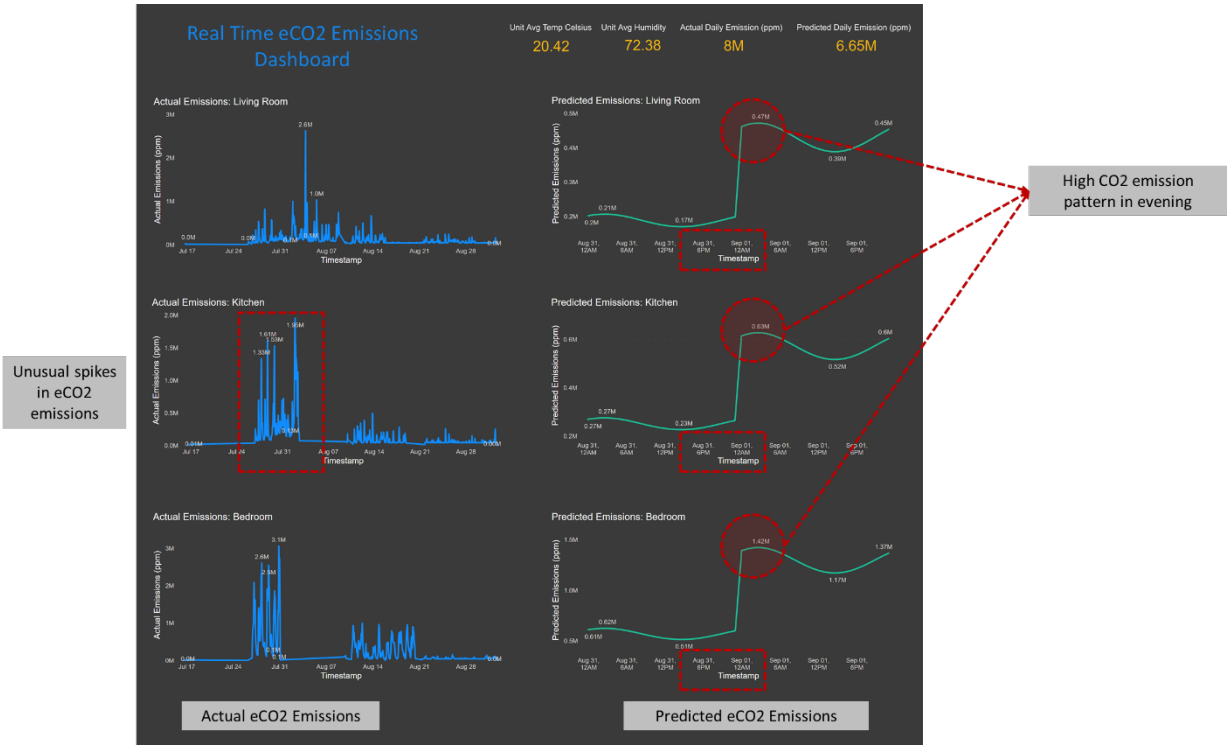


Figure 23. Real-time and Predicted Carbon Emission Dashboard

An interesting point to note about actual emissions is that they start rising during the evening hours, reach peak emissions close to midnight, steadily decline through the rest of the night and early morning hours before plateauing during afternoon. Although the values of the predicted and actual emissions may differ, but this trend (or behaviour) in both the graphs is more or less consistent. This can be attributed to the occupant's morning to evening work schedule, profoundly indicating higher utilization of space during the evening, and reflecting, need for a more effective ventilation strategy during that period to maintain indoor air quality and reduce emissions.

The Scatter plots in Figure 24 show the improvement in the prediction model as compared to the baseline model. The prediction results indicate the enhancement of the trained model as reported by the mean square error and the distribution of the prediction values across the residual plot. There are certain abnormal spikes noticeable in real time eCO_2 emissions data which can have a major impact on the model reliability and accuracy, requiring further data collection over a longitudinal time horizon of a year.

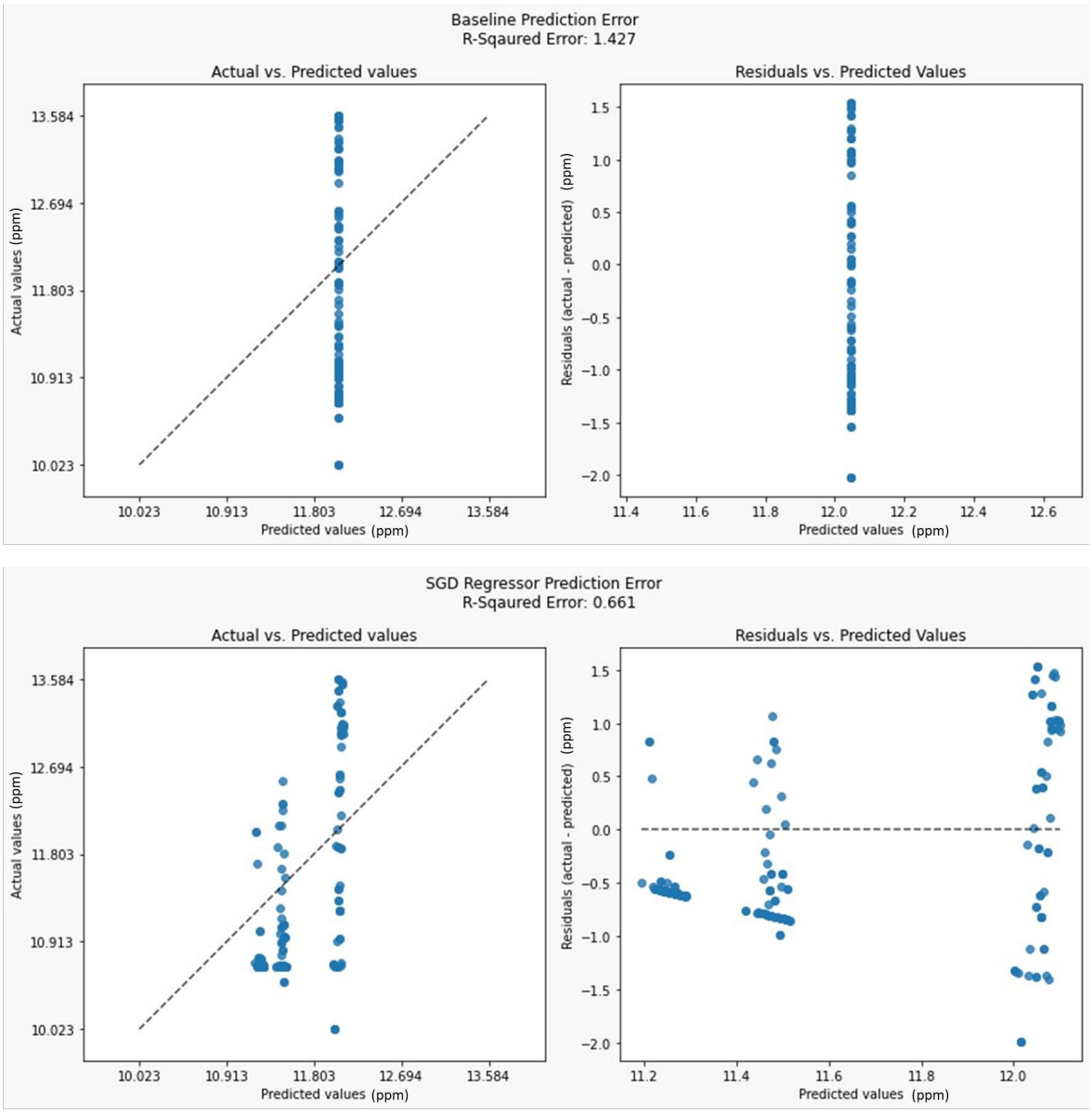


Figure 24. Comparison of the Baseline model (above) and Prediction model (below)

6 Discussion, Significance, and practical implications

The advancements across the ICT sector have been a major driving force behind industry 4.0 transformation, with IoT being one of the key technologies enabling fusion of physical and virtual entities across the cyber-physical system.

The proposed solution development of the DTI architecture reflects a basic integration of 4IR technologies, IoT devices, 3D BIM Model, Laser scanning, Cloud computing and Big Data Analytics, indicating the most essential components for its conceptualization for existing assets in accordance with the findings of Deng, Menassa and Kamat [9] and Khajavi, Motlagh, Jaribion, Werner and Holmström [62], further also presenting the aspirations as well as technological readiness for overcoming limitations noted by Shahzad, Shafiq, Douglas and Kassem [28]. The results of the study demonstrates the potentials of technological advancements for ensuring sustainability and maximizing operational performance of assets, forming the basis of the construction 4.0 framework, as highlighted by various researchers, Tang, Shelden, Eastman, Pishdad-Bozorgi and Gao [11] and Begić and Galić [34].

The most important aspect of the IoT system component for a digital twin, is selection of reliable sensors to receive the required data, therefore identifying this as a crucial step in conjunction to the Assessment and Need of BIM process, defining the basis of this solution development and its purpose, that is to monitor indoor air quality and predict emissions from a housing unit.

Further this study adopts a 3 tier IoT architecture, in accordance with various researches based on smart buildings [47, 105, 106], involving sensors (SGP30 and AHT20) collecting environmental information of space, exchanging this information from IoT device (ESP 32 microcontroller) to the cloud platform through the WiFi network for its further application for visualization and data analysis across the dashboard.

The DTI architecture for the proposed solution includes the 3D Digital Model Layer for capturing the existing assets static data and integrating it with the BIM Model to depict the real-time information, differentiating it from the IoT architecture proposed for smart buildings and smart cities by Moreno, Terroso-Sáenz, González-Vidal, Valdés-Vela, Skarmeta, Zamora and Chang [107].

The developed DTI solution incorporates a cloud platform to store the 3D BIM Model (in glTF format) and sensor data distinctively, further, appending information from IoT devices to the Model for visualization through a REST API to integrate resources in cloud, resolving the interoperability issues for real-time data and model integration, in comparison to previous studies that mostly utilized BIM Tools' API such as Revit API to link data stored in relational database to the 3D BIM Model [108, 109].

This research acknowledges the importance of 3D BIM Model as an essential component of the digital twin, in accordance with the studies by Kaewunruen, Rungskunroch and Welsh [110], Kaewunruen, Sresakoolchai, Ma and Phil-Ebosie [111] Kaewunruen, Rungskunroch and Welsh [110], which is developed through laser scanning of existing asset. The visualization across the 3D Model demonstrated through the case study implementation (Section 5), illustrates an immersive and user-friendly interface, validating the flow of environmental information of the space collected by IoT devices, i.e., temperature, humidity, TVOC and eCO_2, updating real-time changes through coloured mapping, providing better observational insights for asset managers to make data-driven decisions and efficient facility management.

Moreover, the study is a further enhancement of the recommendations identified by Villa, Naticchia, Bruno, Aliev, Piantanida and Antonelli [57] and Tang, Shelden, Eastman, Pishdad-Bozorgi and Gao [11] for developing a digital twin solution, facilitating integration of time-series sensor data in cloud with Azure machine learning to predict emissions trends and anomalies from existing assets, based on which asset managers can make more effective retrofitting decisions in future. The developed dashboard in Power Bi, also enables comparison of predicted and actual emissions acquired by sensors, analysing assets performance in real-time, noting the similarity in the peak eCO_2 concentrations from bedroom during evening period, important information to implement efficient ventilation strategy during retrofitting.

Furthermore, it implements an incremental learning method for model training to update itself once every 2 weeks based on received data, where the mean squared error of the last trained machine learning model was noted to be 0.6, as compared to the baseline model error which was 1.049, indicating the improved predictability of eCO_2 emissions, increasing

efficiency and scalability of the solution, allowing a seamless integration between IoT, BIM and Artificial intelligence within one platform.

Although research in domain of Digital twin is progressing in manufacturing industry, however it is still in nascent stages with limited proven case analysis across the AEC-FM sector [28]. Hence, the presented solution development focusses on a high-level architecture and functionality, tailored to AEC-FM industry, for enabling a complex heterogenous computing environment, while also validating its effectiveness in a practical setting for widespread industry adoption.

The design and development of an artefact in Design Science Research strategy, contributes to the technology body of knowledge as well as extends its practicality by delivering actionable innovative solution to resolve real world problems [89], as evident in this research through the proposed DTI Architecture solution and its corresponding case study validation.

However, the main innovation contribution of digital twin-enabled systems is in the definition of digital technologies that may reduce the performance gap between expected and actual building behaviour, which is essential for the realisation of net-zero buildings.

The digital twin's ability to predict eCO_2 based on the stored historical data acquired by sensors, presents an opportunity to inform the occupants and facility managers of the behaviour of assets which can be further linked to climatic conditions, to implement better retrofitting strategies for ventilation systems of residences ensuring resilience to Sick building syndrome (SBS) and healthy indoor environment as well as to automate its functionality, subsequently regulating the energy consumption pattern.

Although, the implemented Digital twin in this research is more relative to a digital shadow, based on its capabilities, while its framework is scalable to enable bi-directional flow of data from the twin to its physical entity, which can be evaluated in future studies at a wider scale.

7 Research novelty and originality

Digital twins enable operation of high-performance and efficient buildings in the built environment, an emerging component in facility lifecycle management [112]. However, the literature analysis indicates a notable gap in its implementation across sustainability

management of existing structures with few studies focussed on university campus and office environment [14, 27] use cases while others extending to an urban scale [65], to analyse the user interactive outcome of the solution and lacking an in depth methodology to replicate the same for the industry.

With this in mind, the research proposes an enhanced bottom-up solution with open-source codes to promote other novice developers to build on this system, aiming to overcome the technological and functional limitations for integrating BIM, IoT and Artificial intelligence in one platform. Moreover, the case study demonstration addresses the brownfield housing sector in UK and specifically Northern Ireland, providing a different perspective for advancement of digital twin in the industry as well as an initiation towards achieving the realization of efficient net-zero building strategies, interlinked to indoor air quality and ventilation of spaces.

Most research on digital twins for existing assets is limited to static information, such as simulating energy consumption data across the Scan to BIM geometrical Models, to offer net-zero retrofitting strategies [26, 110, 113]. While this paper emphasizes on the integration of real-time environmental data across the 3D BIM Model of existing asset, in addition further developing a solution to utilize the stored historic data to forecast future eCO_2 emissions through machine learning trained model, to bridge the gap between as-designed and as-built asset.

Besides, the proposed solution utilizes the RealEstateCore Ontology to replicate the physical asset and its environment as a digital twin graph, in an effort to streamline the asset information management based on industry standards, overcoming the interoperability issues for exchange and storage of real-time information [15, 57].

With exception of previous IAQ studies, which primarily presented statistical information [98, 114, 115] that may have required technical knowledge for interpretation, the proposed digital twin solution automates the back-end processes to display alerts and critical information through shaded rendering across the 3D BIM Model in real-time, enhancing the usability of the system as well as provisioning an immersive and user-friendly dashboard for both occupants and facility managers.

To conclude, the proposed DTI architecture can be implemented across any project type, and is not limited by any specific sensor chosen, hence facilitating a flexible and reusable solution for the industry, to develop digital twins for existing assets.

8 Conclusion

The study presents a fundamental DTI architecture for implementation on existing assets, tested on a case study, selected to represent a small-scale residential unit, to illustrate its validity and workability, providing an immersive, user-friendly interface for visualizing indoor air quality of a space in real-time across the 3D BIM Model with alerts integrated for observation of critical information. In addition, it simultaneously utilized artificial intelligence to predict future emissions based on received time-series data from installed devices, by implementing a machine learning algorithm, the results of which were analysed in a dashboard to observe trends in pattern, indicating the need for a more efficient ventilation strategy for the space.

The detailed architecture of this solution is flexible, facilitating incorporation of further devices and data, reinstating its reusability for industry adoption, whilst also providing a foundation for future researchers to build upon this study to further develop enhanced digital twin solutions for existing assets.

The proposed solution has significant implications in three aspects: (1) Widespread digitization of existing assets to monitor their condition in real-time, (2) Enabling facility managers to monitor asset performance based on established net zero targets, (3) entailing a collaborative platform for various stakeholders to leverage information to implement better renovation strategies in future to reduce energy consumption as well as emissions.

This study unveils new prospects for digitising existing housing industry assets through integration of emerging technologies to develop a digital twin, which shall in turn facilitate effective asset and sustainability management. The incorporation of digital twins of housing assets at a national level, such as the on-going efforts of CDBB for developing a NDT for UK, a macro scale development, shall allow policy makers, to recommend comprehensive and stringent data-driven strategies to achieve 2030 targets of reduced emissions, based on the performance of existing assets.

Despite the positive implications, there is a further need for interdisciplinary collaboration with ICT engineers and AEC-FM practitioners, to integrate IFC with RealEstateCore Ontology to build seamless relationship across data from BIM Models, CoBie and IoT sensors, streamlining the data exchange standards for the industry.

Nevertheless, the study identifies certain limitations for the solution development, limited to implementation of Digital shadow, capable of real-time diagnosis and condition monitoring, but provisioning extension of this architectural framework to enable a comprehensive Digital twin further enhancing its capabilities for achieving Net zero as part of future research, supporting predictive maintenance, what if analysis to test scenarios and autonomous decision making, allowing scope for an interactive involvement of owners and inhabitants to implement smart building automation in residences, by deploying multiple IIoT devices and integration of heterogenous information, such as energy use intensity, water consumption, waste generation.

Moreover, the study does not explore the organizational effectiveness of implementation of digital twin for asset management, another area for future research to motivate the various project participants for its adoption and increasing data transparency, breaking the industry silos.

In future, an empirical evaluation of the proposed system architecture through implementation on a broader scale and demonstration to industry practitioners, shall allow to further validate, investigate, and enhance its wider adoption across the housing sector, eliminating the potential challenges.

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